



Psychology in America: The Early Years

PSYCHOLOGY FIRST BECAME AN ACADEMIC SCIENTIFIC DISCIPLINE IN Germany because Germany developed the modern university system that enabled Wundt and his colleagues to create an experimental psychology grounded upon the earlier success of experimental physiology. However, psychology developed faster institutionally in America than it did in Germany (Ash, 1980; Danziger, 1979). By the end of the 19th century, American psychology had an active professional organization, the American Psychological Association, founded in 1892; journals devoted to general, experimental and applied psychology, such as the *American Journal of Psychology*, *Pedagogical Seminary*, *Psychological Review*, *Psychological Index*, and *Psychological Monographs*; and a substantial academic presence within the American university system. By 1892, America had more and better laboratories than Germany. Formal research laboratories were established at the University of Chicago, Clark University, Columbia University, Cornell, Harvard, Pennsylvania, Wisconsin, and Yale; and demonstration facilities for teaching and training were available at Brown, Catholic University of America, Indiana, Kansas, Michigan, Nebraska, Illinois, Iowa, and Wellesley (Hale, 1980; O'Donnell, 1985). By 1904, there were 49 laboratories, 169 members of the APA, and 62 institutions offering three or more courses in psychology; in addition, psychology ranked fourth in the sciences with respect to the number of PhDs awarded (Camfield, 1973; Miner, 1904). By 1913, the year that Watson published his behaviorist manifesto "Psychology as the Behaviorist Views It," America had surpassed Germany in research publications (Cattell, 1917).

Much the same was true of the academic institutionalization of other social sciences in the early 20th century. Émile Durkheim (1858–1917) introduced sociology in France at the end of the 19th century, but it made slow progress, and by 1930 there were only three chairs in sociology. Yet by 1910 there were about 50 full-time professors of sociology in America, and the subject was taught at around 400 colleges (Smith, 1997).

In contrast to the often reactionary response to attempts to introduce psychology in Britain and Europe, in America it was generally accepted by the public,

government, and university administrators as a legitimate intellectual and practical pursuit, with much to offer the academy and society at large. Psychological questions were matters of common interest in the increasingly secular, industrial, and urban American culture, as evidenced by the popular appeal of phrenology and mesmerism. In 1850 the Reverend John Bovee Dods (1795–1872) gave a series of lectures on “electrical psychology” in the Hall of Representatives in Washington, DC, at the invitation of state senators such as Sam Houston, Henry Clay, and Daniel Webster. His published version of these lectures, *Electrical Psychology* (1850), became a national best seller (Reed, 1997). The democratic culture and progressive social values of late-19th-century America made it fertile ground for the development of the new social sciences, including psychology. And American psychologists were not hesitant in the promotion of their new discipline, as they strove for public and administrative support.

American psychology was originally based upon the German model of laboratory research, but those Americans who trained with Wundt adapted the new science to their own idiosyncratic interests, as well as local institutional and social demands. Most psychology departments were established in conjunction with laboratories and conducted practical teaching demonstrations of experiments in psychophysics, reaction time, and sensory and perceptual processes throughout the early decades of the 20th century. However, American psychologists did not restrict themselves to research in physiological psychology, but directed their scientific interests to a wide range of topics in applied psychology, such as education and industry. Indeed, perhaps the most distinctive feature of early American psychology was its eclecticism, which promoted the early growth of scientific approaches to social and psychological problems that eventually developed into full-blown subdisciplines of psychology, such as educational, industrial, clinical, social, developmental, forensic, and personality psychology.

PSYCHOLOGY AND THE DEVELOPMENT OF THE AMERICAN UNIVERSITY

Prior to the U.S. Civil War, colleges of higher education were largely devoted to the training of ministers and the intellectual elite. They were also rather small affairs. Although already nearly 200 years old, Harvard had fewer than 20 faculty in the 1850s. Much emphasis was placed upon the classics, mathematics, and logic as a means of fostering mental discipline. The educational goal of most early-19th-century colleges was to organize natural, moral, and religious knowledge into a coherent system.

Moral philosophy (or “moral science”) was based upon the Scottish common sense psychology of Reid and Stewart and was taught to seniors as the capstone of their liberal education. It affirmed the reliability of perception and intuition in

the pursuit of natural, moral, and religious knowledge and maintained the traditional dualist distinction between immaterial mind and material body. Scottish Presbyterian émigrés such as James McCosh (1811–1894), the president of Princeton, and Noah Porter (1811–1892), the president of Yale, promoted this form of moral philosophy in America. McCosh was the author of *The Scottish Philosophy* (1875) and *Psychology: The Cognitive Powers* (1886); Porter's *The Human Intellect: With an Introduction Upon Psychology and the Soul* (1868) was based upon Dugald Stewart's *Elements of the Philosophy of the Human Mind* (1792).

Moral philosophy was not antagonistic to science. On the contrary, moral philosophers committed to Scottish common sense psychology were philosophical realists who argued that the immaterial mind can have genuine knowledge of the material world through veridical perception and the employment of scientific methods. The sciences themselves expanded considerably within the American educational system in the middle decades of the 19th century, when colleges upgraded their teaching in mathematics, physics, and astronomy and added subjects like chemistry, geology, and biology. By the late 19th century, scientific training, including the laboratory methods of the new scientific psychology, came to be accepted as a legitimate alternative to the classics as a means of instilling mental discipline (O'Donnell, 1985).

The Morrill Act of 1862 promoted the development of state universities by awarding land and grants to state-founded institutions of research and instruction, especially in new technological areas such as mining, agriculture, and industry (although some state universities predated the act, such as the precursors of the universities of Georgia, Iowa, Michigan, and North Carolina). Competition for resources and prestige between these state institutions, which were usually governed by boards of local businessmen and politicians, led to their inevitable expansion, including the offering of a wider range of academic subjects.

Some of the major American universities founded in the late 19th century were effectively the creation of American business, and the emerging social scientific disciplines were to an almost unhealthy degree the creatures of American capitalism (Manicas, 1987). Cornell University, which opened in 1865, was founded by a huge financial bequest from the businessman Ezra Cornell (1807–1874), who insisted on control of administrative matters at the university. Andrew D. White (1832–1918), the first president of Cornell, was a crusading advocate of secular and scientific approaches to education. He vigorously opposed the historical influence of religion on the universities and published *A History of the Warfare of Science With Theology in Christendom* in 1896. He introduced a new “liberal arts” curriculum that was no longer restricted to the classics and other formal disciplines, but included natural and social sciences (including psychology), literature, art, and history.

Charles Eliot (1834–1926), appointed president of Harvard in 1869, became a champion of academic professionalism. While he continued to promote the

virtues of a general liberal education, he also recognized the need for academic specialization attuned to the demands of an industrial society. He expanded the curriculum and introduced the then novel idea of an elective system of course offerings in place of the older and more rigid system of required courses. When philosophy suffered as a result, Eliot allowed William James to introduce a course on the new physiological psychology. By 1878 Eliot had doubled the instructional staff, recruiting faculty on the basis of their specialized qualifications rather than (as previously) their character and alumni status. As the academic market became more competitive, Eliot increased faculty salaries and welcomed curricular innovations such as James's course in psychology as a means of increasing enrollment to fund expansion (O'Donnell, 1985).

Academic competition accelerated as a consequence of the founding of Johns Hopkins University in 1876, the first exclusively graduate school in America, at the bequest of the Baltimore financier Johns Hopkins. On the instructions of the businessmen-trustees, Daniel Coit Gilman (1831–1908), the first president of Hopkins, tried to create from scratch a national center of research and graduate education oriented to science and modeled upon the ideals of the German university. Gilman attracted faculty to the new university by introducing lighter teaching loads and financial support for research (including the provision of research assistants). He attracted graduate students by introducing graduate fellowships, and between 1876 and 1880 graduate enrollments at Harvard, Yale, and Cornell dropped precipitously. The older universities were forced to compete by refurbishing their graduate programs, creating the “precondition for the American development of psychology as a research science and an academic profession” (O'Donnell, 1985, p. 115). In 1884 G. Stanley Hall (1844–1924) created the first laboratory and PhD program in psychology in America at Johns Hopkins, although he virtually dismantled the program when he accepted the presidency of Clark University in 1888, taking his staff, students, journal, and laboratory apparatus with him.

Perhaps the most famous academic product of American capitalism was the University of Chicago, founded by John D. Rockefeller in 1890. Designed to bring prestige to the new metropolis and address its pressing social problems (such as urban growth, labor conflicts, and crime, themselves arguably the product of runaway capitalism), Chicago used its financial muscle to raid other universities for talent and quickly became a recognized center of excellence in a variety of disciplines, including psychology. Distinguished psychologists associated with the university included John Dewey (1859–1952), James Rowland Angell (1869–1949), John B. Watson, Harvey Carr (1873–1954), George Herbert Mead (1863–1931), Helen Woolley (1874–1947), Walter V. Bingham (1880–1952), and Walter S. Hunter (1889–1956).

Around this time a number of colleges for women were also founded. Vassar College, founded in 1861, provided the intellectual stimulus for a number of early American women psychologists, such as Lillien Martin and Margaret Washburn

(1871–1939). Martin, a former teacher, studied with Müller. Washburn studied with James McKeen Cattell and Edward B. Titchener and was the first woman to attain a PhD degree in psychology (at Cornell). She returned to a teaching position at Vassar, where she had a long and distinguished career, being elected president of the APA in 1921 and to the American Academy of Sciences in 1931 (Bolles, 1993).

The new private universities began to incorporate the German model of research disciplines and PhD certification of university teachers, as the older ones scrambled to revise their goals and curricula in response to the new conditions and the new state colleges and universities developed and expanded their own. This created ideal conditions for institutional investment in a new discipline like psychology, identified as one of the successful 19th-century products of the German university system.

This academic expansion, especially in new disciplines such as psychology, created a critical demand for qualified PhDs. This was a demand that the nascent American system was initially unable to satisfy, since few colleges outside the Ivy League were able to grant the PhD degree. It was for this reason that many aspiring American educators in the new social scientific disciplines went to Europe for professional certification. Early American psychologists were attracted to Wundt's program largely because of the professional qualification it offered and the advantages it provided them in securing positions and advancement at the more prestigious academic institutions back home. James Mark Baldwin (1861–1934), who founded laboratories at Princeton and Toronto, frankly admitted that he spent time in Leipzig only "to secure a distinct advantage in the professional race at home" (Baldwin, 1926, 1, p. 35). Much the same was true of many of the institutional founders of other social sciences in America at the end of the 19th century. Albion Small (1854–1926), one of the founders of sociology at the University of Chicago, studied at the Universities of Berlin and Leipzig, and Franz Boas (1858–1942), the founder of anthropology at Columbia University, studied at the Universities of Heidelberg, Bonn, and Keil (Manicas, 1987).

American psychologists who returned from Leipzig and other German universities reproduced the institutional structures of the German university, notably the laboratories and research seminars associated with the new scientific disciplines (Veysey, 1965). Yet few American psychologists were imbued with the German commitment to scientific research for its own sake, and fewer still pursued research careers devoted to the experimental programs pursued in German laboratories. During the early decades of American psychology, graduate students were trained in traditional laboratory techniques, but more as a means of instilling a spirit of scientific rigor than as a preparation for research careers in physiological psychology (O'Donnell, 1985). Laboratory training served as the mark of scientific authority and status for academic psychologists, distinguishing "expert" from amateur psychologists such as traditional moral philosophers, physiologists, physicians, mesmerists, and spiritualists.

American psychologists regularly invoked the laboratory-based nature of their fledgling discipline as they strove to secure the autonomy and advancement of their discipline in the years prior to the First World War, citing experimental rigor as the basis of the objectivity of their new science (O'Donnell, 1985). However, some did question the identification of psychology with laboratory science. Joseph Jastrow (1863–1944), the first person to gain a PhD in psychology in America (at Johns Hopkins), warned against equating scientific psychology with experimentalism (Jastrow, 1887). William James, the first to introduce courses on the new physiological psychology, soon became skeptical of the utility of “brass instrument” psychology and championed a more practical and applied approach. Yet, with few exceptions, American psychologists simply adjusted their own laboratory-inspired scientific attitudes to whatever topics happened to interest them or to whatever opportunities for research happened to arise in their varied careers.

Although American psychologists returned with only the academic and experimental skeleton of German psychology, they set about reproducing it with such speed and efficiency that American psychology quickly overcame German psychology. By the turn of the century, dozens of laboratories and psychology programs were established at both Ivy League and state universities, many by Wundt's former students. Laboratories and programs in psychology were also established at smaller private colleges, including women's colleges. Mary Calkins founded a laboratory at Wellesley College in 1891, and Lillie Williams founded a laboratory at the State Normal School at Trenton in New Jersey in 1892 (Bolles, 1983).

By 1917, there were 307 members of the APA (84 percent with a PhD), 74 laboratories, and 35 separate departments of psychology, with psychology represented in 122 institutions (Camfield, 1973, based upon Cattell, 1917). The number of American psychologists listed in the British publication *Who's Who in Science* that year equaled the combined number in Germany, France, and Britain. By 1929, there were around 1,000 psychologists in America, representing over 300 institutions—outnumbering the total number of academic psychologists in Europe, if not the rest of the world (Cattell, 1929).

The Success of Psychology

There were a number of reasons for the success of psychology as an academic discipline in America. While aspiring psychologists in Germany and Britain had to battle reactionary philosophers and conservative administrators, American philosophers generally welcomed the **new psychology**, as experimental psychology was called, and the pragmatically oriented businessmen and politicians who directed American universities were hospitable to the new discipline and persuaded of its great social promise (albeit much exaggerated by its promoters).

The new psychology was not generally perceived as a threat to the traditional form of moral philosophy based upon Scottish common sense psychology. What was perceived as a threat was the materialism and mechanism of Darwin's theory of evolution by natural selection and the reflexive sensory-motor theory of the nervous system. Like most scientists, moral philosophers accepted the fact of evolution, but were opposed to Darwin's claim that evolution could be explained as the mechanical outcome of biological variation and natural selection, without reference to divine purpose and design. They condemned his treatment of human psychology and behavior as strongly continuous with that of animals, since, like Descartes, they saw the denial of the distinctiveness of human rationality as a threat to freedom and moral responsibility. They also followed the proponents of the sensory-motor theory of the nervous system in supposing that the theory implied that cognition and consciousness are merely epiphenomenal by-products of the reflexive neurophysiological processes directly responsible for all animal and human behavior.

Traditional moral philosophers embraced Wundt's physiological psychology as an intellectual bastion against materialism and epiphenomenalism, since Wundt claimed that psychological principles could not be reduced to physiological principles and insisted upon the voluntary nature of "psychic causality." American moral philosophers were quite happy to subject consciousness to the experimental control of the laboratory, because they saw it as an affirmation of the reality and autonomy of consciousness against the materialist reductionism of evolutionary theory and experimental physiology.

James McCosh, a major interpreter of Scottish common sense psychology, who defined psychology as the "science of the soul" (1886, p. 1), introduced courses on the new psychology at Princeton. George Trumbull Ladd (1842–1921), who took over the teaching of psychology at Yale from Noah Porter, introduced the new psychology and promoted it in *Elements of Physiological Psychology* (1887), a work that combined Wundt's physiological psychology with traditional elements of Scottish common sense psychology. Ladd maintained that Wundt's physiological psychology, which transcended its physiological origins, had rescued mind from the materialism and epiphenomenalism of the physiologists, who denied "the reality, unity and possibility of a permanent existence of the human mind" and reduced it to "a stream of mechanically associated 'epiphenomena,' thrown off from the molecular machinery of the cerebral hemispheres" (1895, p. x).

Academic administrators recognized the threat posed to traditional moral philosophy by developments in experimental physiology. The overseers of Harvard University, for example, complained that the philosophical neglect of "the physical side of mental phenomena has had the natural effect of exaggerating the importance of materialist views" (O'Donnell, 1985, p. 62). This led Princeton's McCosh to proclaim that moral philosophers ("metaphysicians") should engage

the implications of experimental physiology:

The metaphysician must enter the physiological field. He must, if he can, conduct researches; he must at least master the ascertained facts. He must not give up the study of the nervous system and brain to those who cannot comprehend anything beyond what can be made patent to the senses or disclosed to the microscope.

—(McCosh, 1875/1980, p. 458)

Consequently, pioneers of the new psychology in America such as William James and G. Stanley Hall had little difficulty in representing the new psychology to the presidents and trustees of Harvard and Johns Hopkins as antithetical to crude materialism and mechanism (Leary, 1987).

Philosophy and Psychology

Some philosophers resisted the development of scientific psychology. Frances Bowen, the Alford Professor of moral philosophy at Harvard, vigorously but ineffectively protested the listing of James's 1877 course on "Physiological Psychology" in the department of philosophy (Leary, 1987). However, many others supported the development of psychology as a scientific discipline, and philosophers such as James, Dewey, and Hall played a major role in establishing psychology as an autonomous scientific discipline. Eventually American psychologists came to resent the constraining influence of philosophy and campaigned for autonomous chairs and departments of psychology, but in the early years philosophers and psychologists saw the new psychology as a means of revitalizing traditional philosophy.

In a very real sense it did. The autonomous academic disciplines of philosophy and psychology emerged at around the same time at the end of the 19th century. American academic psychology developed institutionally by sundering its connections with philosophy, but the disciplinary identity of late-19th- and early-20th-century academic philosophy was itself developed in reaction to the emerging discipline of scientific psychology. Anglo-American **analytic philosophy**, based upon the logic of Gottlob Frege (1848–1925) and Bertrand Russell (1872–1970), and Continental **phenomenological philosophy**, based upon the phenomenology of Brentano and Husserl, were developed in reaction to **psychologism**, the theory that logical and conceptual relations can be treated as naturalistic psychological "laws of thought." This was the theory advanced by John Stuart Mill and John Venn (1834–1923) and accepted by many practitioners of the new psychology. In contrast, analytic philosophy defined itself as a discipline concerned with the conceptual analysis of abstract systems of thought (based upon the new forms of symbolic logic developed by Frege and Russell), and phenomenological philosophy defined itself as a discipline concerned with the abstract essence of thought. Thus both psychology and philosophy, which shared a common prehistory, developed

in the 20th century as autonomous academic disciplines with the emergence of the new psychology originally developed within philosophy departments at the end of the 19th century (Reed, 1997). The other distinctive form of 20th-century philosophy, American **pragmatist philosophy**, was an offshoot of evolutionary theory and American functionalist psychology (O'Donnell, 1985).

Philosophy certainly developed from psychology at the level of professional associations. The American Psychological Association was founded in 1892, 10 years before the American Philosophical Association. In the early years, many philosophers joined the APA and presented papers at the annual convention. This eventually created a problem, since at the 1896 and 1898 meetings (at Harvard and Columbia) there were more papers on philosophical topics than on psychological ones. This led to some minor friction, leading to calls for the creation of a separate philosophical association, originally conceived of as a division of the American Psychological Association. The problem was resolved when philosophers meeting in Kansas City formed the Western Philosophical Association in January 1900. Philosophers on the East Coast followed by creating the Eastern Philosophical Association in 1901. The two associations later amalgamated, and the American Philosophical Association now includes Eastern, Central, and Western divisions (Sokal, 1992). The American Psychological and Philosophical Associations continued to meet together for a number of years, a practice nowadays maintained only by the Southern Society of Philosophy and Psychology.

From the point of view of traditional moral philosophy, the acceptance of the new psychology was a mixed blessing. It proved to be a Trojan horse bearing the progenitors of later forms of materialism and epiphenomenalism. Laboratory-based physiological psychology gave way to a functional psychology avowedly inspired by evolutionary theory and eventually to a behaviorist psychology that equated consciousness with the outdated notion of an immaterial soul.

Applied Psychology

The pragmatically oriented presidents and boards of trustees of American universities were much more enthusiastic about the prospects of psychology than were their European counterparts. And if they had any doubts, American psychologists were extremely skillful in presenting the new psychology not only as compatible with religion and traditional moral philosophy, but also as a great benefit to society.

If American psychologists did not generally deliver on their grandiose visions of the potential of psychology to improve education, industry, and mental health, they were masterful in their promotion of the social utility of the discipline. Indeed, it might be fairly said that the promotion of psychology as a discipline was their greatest applied psychological achievement. From the very beginning, James, Hall, Dewey, and Cattell presented psychology as offering an applied scientific approach to human psychological and behavioral problems—as a discipline

of immediate practical relevance to educators, businessmen, medical practitioners, advertisers, and penal administrators.

From the early years, American psychology entered into a fertile developmental relationship with American education. As primary and secondary education expanded dramatically in the later decades of the 19th century and early decades of the 20th century, American psychologists established their intellectual authority with respect to the training of the new generation of teachers and the progressive reform of education. Psychologists came to be revered by educators as representing the scientific hope and promise of their discipline. Although psychology had perhaps little of concrete value to offer the educators, the early focus on education powerfully shaped the course of the later development of psychology, including the intensive focus on theories of learning in the first half of the 20th century (Leary, 1987).

The rhetorical emphasis on the potential of psychology to alleviate human suffering and improve the human condition through scientifically informed social intervention was entirely concordant with the evangelical goals of Scottish common sense moral philosophers (Fuchs, 2000). Many of the early pioneers of American psychology inherited the crusading mantle of traditional moral philosophers, and it is probably no accident that many originally considered careers in the ministry. James considered such a career, and Ladd, Hall, Walter D. Scott, and Watson originally trained for the ministry.

The earliest and originally most influential centers of scientific psychology were at Harvard, Johns Hopkins, Clark, Cornell, Chicago, and Columbia. Yet in the early decades of American psychology, literally hundreds of psychologists were working on just about every aspect of psychology at American universities and colleges, where Wundt was taught along with Bain, Spencer, Darwin, Galton, and Morgan, and James, Ladd, and Dewey. While the 20th-century history of American psychology can be fairly represented in terms of a historical progression from experimental psychology to a more eclectic functional psychology and from the varieties of behaviorism to the cognitive revolution, one does well to remember that these movements represented only the major orienting currents of the discipline at particular points in time and not the complex undercurrents and subdisciplinary developments upon which they supervened.

JAMES AND MÜNSTERBERG AT HARVARD

William James

William James holds a peculiar position in the history of American psychology. He anticipated many of the later developments of scientific psychology, but did not found any distinctive school and had few theoretical disciples. Yet he scores consistently high in surveys of influential figures in the development of

American psychology. Cattell's 1903 poll of leading psychologists ranked him first in influence, and later polls invariably place him in the top 10. Although present day psychologists admire the encyclopedic breadth of his interests, many of his contemporaries found him too eclectic and strongly disapproved of his interests in spiritualism and free will.

James himself was ambivalent about psychology. Originally enthusiastic about the prospects for a scientific psychology, he hated laboratory work and was not much impressed by the work of German pioneers such as Fechner, Wundt, and Müller. He was respectfully critical of Wundt, but cruelly dismissive of most of the others. He described Fechner's psychophysics as destined to amount to "nothing" and Müller's experimental work as "brutal." In later years James became disillusioned with what he called the "nasty little subject," but worked conscientiously throughout his life to promote its development. He founded a demonstration laboratory at Harvard and encouraged interested students to pursue a career in the new psychology, correctly predicting that departments of philosophy would soon have "a number of vacant places calling for their peculiar capacity" (1876, p. 179).

While he championed psychology as a laboratory science capable of safeguarding traditional moral philosophy and religion from the materialism and epiphenomenalism of reductive physiology, he came to disparage "the new prism, pendulum and chronograph-philosophers." From the beginning he emphasized that psychology was a positivist science concerned with "practical prediction and control," with much of value to offer to society:

We live surrounded by an enormous body of persons who are most definitely interested in the control of states of mind, and incessantly craving for a sort of psychological science that will teach them how to act. What every educator, every jail-warden, every doctor, every clergyman, every asylum-superintendent, asks of psychology is practical rules. Such men care little or nothing about the ultimate philosophical grounds of mental phenomena, but they do care immensely about improving the ideas, dispositions and conduct of the particular individuals in their charge.

—(1892a, p. 151)

James was born in Astor House, an opulent New York hotel, into a wealthy and cultured family. He had three brothers (one of whom was the novelist Henry James) and one sister. He was educated in Europe and America, in a stimulating social and intellectual environment that included personal acquaintance with many of the leading intellectuals of his day, such as Ralph Waldo Emerson, Henry David Thoreau, Thomas Carlyle, and John Stuart Mill. His father encouraged him to be a scientist, although James originally explored a career in art and studied with William Hunt in Newport, Rhode Island. He abandoned art in 1861 to study chemistry and physiology at Harvard (as others went off to the Civil War), switching to medicine in 1864. He found he had little interest or native aptitude for any

of these subjects, and a zoological expedition to the Amazon in 1865 convinced him he had none for biology either. James contracted smallpox, and his health, never good, deteriorated on his return to Harvard. He suffered backache, poor eyesight, insomnia, and depression (which may have been psychosomatic in origin, at least according to James).

A compulsive traveler all his life, he took himself off to Germany in 1867, hoping that he might relieve his back pain by taking mineral baths at local spas. There he visited Fechner, Helmholtz, du Bois-Reymond, and Wundt, and came to believe that “perhaps the time has come for psychology to begin to be a science” (1920, 1, p. 118). Returning to America, James completed his medical degree in 1869 and immediately resolved never to practice medicine. His health deteriorated further, and he became depressed and suicidal. In characteristic fashion, James claimed to have cured his depression by convincing himself of his own free will, through his act of coming to believe in free will. His first act of free will was “to believe in free will” (1920, 1, p. 147).

James was offered the post of instructor in physiology at Harvard medical school in 1872. In 1875 he managed to persuade Charles Eliot, the president of Harvard, to let him introduce a course on “The Relations Between Physiology and Psychology.” That same year he managed to get money from Harvard to set up a laboratory for experimental demonstrations. “Laboratory” is probably somewhat of an exaggeration, since as Hall later pointed out, it included no more than “a metronome, a device for whirling a frog, a horopter chart, and one or two bits of apparatus” stored in a stairwell closet (Hall, 1923, p. 218).

James’s course was a great success, and in 1876 he was promoted to the rank of assistant professor of philosophy. He became professor of philosophy in 1885. In 1878 James married Alice Gibbons, whom his father had chosen as his wife (Allen, 1967). This introduced some stability in his life (they had five children), although it did nothing to reduce his recurrent wanderlust. That same year James contracted with Henry Holt to write a book on psychology, which became the *Principles of Psychology*, published in two volumes in 1890. *Psychology: A Briefer Course* was published two years later (1892b).

The *Principles* was also a great success. It was widely read in America and Europe, and by the educated public as well as psychologists. For many years it was the standard psychology textbook in most American and many European universities. However, not all of James’s psychological peers approved of it. Wundt dismissed it as “literature . . . not psychology,” and Edward B. Titchener, the founder of structural psychology in America, condemned it. Later functional and behaviorist psychologists were less than enthusiastic about James’s commitment to introspection, free will, and spiritualism, not to mention his definition of psychology as “the science of mental life” (1890, 1, p. 1).

James seems to have lost interest in psychology after the publication of *Principles*. He arranged for Hugo Münsterberg (1863–1916), a former student of Wundt’s,

to take over his psychology courses and the laboratory at Harvard in 1892 and devoted most of his later years to philosophy (and further travel). However, he continued to fight to maintain Harvard's prominence in psychology (O'Donnell, 1985). He raised over \$4,000 (more than \$90,000 in today's dollars) to purchase laboratory equipment to match the facilities of the newly founded Clark University, since, as he put it to Münsterberg: "We are the best university in America, and we must lead in psychology" (1892c, p. 68). He retired in 1907 and died in 1910 (on returning from a final trip to Europe).

In many ways, James resembled his German friend and colleague Carl Stumpf. Both were skeptical of the pretensions of the new psychology and had little personal taste for laboratory work, but both successfully championed the development of psychology as an experimental and applied discipline in their respective institutions and countries.

The Metaphysical Society In the early 1870s at Harvard, James organized the informal meetings of the **Metaphysical Society**, devoted to the philosophical problems of the day, such as the nature of knowledge, meaning, and truth; the relation between science and religion; and the role of the new psychology in relation to the older common sense psychology of Reid and Stewart. The group included James, the philosophers Chauncey Wright (1830–1875) and Charles Sanders Peirce (1839–1914), and the legal theorist Oliver Wendell Holmes (1809–1894), who later served as chief justice of the Supreme Court. One significant outcome of these meetings was the development of the philosophical view known as **pragmatism**: the view that the adequacy of any theoretical system should be judged by its practical utility. The name derives from Kant's characterization of uncertain beliefs with practical value as pragmatic beliefs (Kant, 1788). Since the members of the society did not think there were (or could be) any certain theoretical beliefs, they held that all theoretical beliefs are pragmatic beliefs: that is, beliefs that are only justifiable by their practical consequences.

They differed concerning the relevant consequences, however. Wright expounded a theory of the selection of beliefs via their practical consequences for an individual during his or her lifetime: a cognitive version of the Spencer-Bain principle. Peirce developed a **pragmatist theory of meaning**, according to which the content of a belief or proposition is specified in terms of its empirical consequences. Pierce also avowed a form of instrumentalism, insofar as he claimed that scientific theories should be evaluated only by their empirical consequences, rather than by reference to their supposed correspondence with theoretical reality.

Although Peirce was the first to develop pragmatism, which he originally called pragmaticism (1905/1982), he was a poor promoter of it during his lifetime. An 1859 Harvard graduate, he never managed to secure a full-time academic job (although he lectured at Johns Hopkins from 1879 to 1884), and worked for most

of his life with the U.S. Coast and Geodesic Survey. James was the one who successfully promoted pragmatism in *The Will to Believe* (1897), *Pragmatism* (1907), and *The Meaning of Truth* (1909). He also developed a **pragmatist theory of truth**, according to which a belief or proposition is true if it “works satisfactorily in the widest sense of the word.” James really did mean “in the widest sense of the word,” because he held that a belief or proposition should be accepted as true if it satisfies our feelings or produces a beneficial effect, such as a religious belief or a belief that mineral baths relieve backache.

Most American psychologists embraced pragmatism in the broad sense that theories should be judged by their practical utility, although few were committed to the specifics of the pragmatist theories of meaning and truth, and most rejected James’s idea that the truth of a theory could be warranted by the satisfaction derived from believing it.

James’s Psychology James was highly critical of Wundt’s experimental analysis of the elemental contents of consciousness and laid much greater stress on the continuity of consciousness. He famously talked of a “stream of consciousness” rather than a “chain” or “train” of conscious states and stressed the selectivity and functionality of consciousness. Yet he criticized a caricature of Wundt, who also recognized these features.

James affirmed the standard account of the association of ideas in terms of similarity and continuity, which he treated as a consequence of elementary laws of neural excitation and succession (although he added secondary principles of association such as vividness and congruity in emotional tone). He followed Hartley, Spencer, Bain, and Morgan in extending associationist principles to the formation of behavioral habits. While he famously held that acquired behavioral habits form the “great fly-wheel of society,” he also followed Darwin in maintaining that much of human and animal behavior is governed by instinct.

James developed an account of emotion that came to be known as the **James-Lange theory of emotion**, published by James in 1884 and independently by the Danish physiologist Carl Georg Lange (1834–1900) in 1885. According to this theory, emotion is not the cause of physiological arousal and behavior, as is commonly supposed, but rather our experience of physiological arousal and behavior. Thus he suggested that we should say that “we feel sorry because we cry, angry because we strike, afraid because we tremble, and not that we cry, strike or tremble because we are sorry, angry or fearful” (1884, p. 190). By this account, our fear of the bear, for example, does not cause us to run away, but is simply our experience of physiological and behavioral responses to our perception of the bear. James’s treatment of emotion as epiphenomenal was a consequence of his commitment to the reflexive sensory-motor theory of the nervous system, which he accepted and defended at length in the *Principles*. Yet although he treated instincts, habits, and emotions as reactive and automatic, he was reluctant to accept that consciousness

and cognition were merely “impotent, paralytic spectators” of reflexive sensory-motor sequences. He retained a causal role for consciousness and cognition in the generation of behavior by embracing a version of Carpenter’s ideomotor theory, according to which selective attention to ideas can determine which automatic ideomotor sequences are activated. James also developed some interesting theoretical distinctions between different aspects of the empirical self, such as the material, social, and spiritual selves.

James’s Influence James functioned as a kind of father figure in the development of American psychology, as an intellectual conduit who reflected 19th-century developments in experimental physiology and evolutionary theory and the pragmatic and functional approaches to psychology that came to dominate in future decades. The demonstration laboratory he instituted at Harvard in conjunction with his first course in psychology in 1875 predated Wundt’s, but was not used for serious research until much later—the first functional American laboratory was created by G. Stanley Hall at Johns Hopkins in 1884 (Hulse & Green, 1986). Scientific psychology would probably have developed in America at the time it did and in the fashion it did even if James had never introduced psychology courses or a demonstration laboratory at Harvard, but his early commitment to the new discipline helped to promote its development, and many individuals were inspired to follow a career in psychology after reading James’s *Principles*.

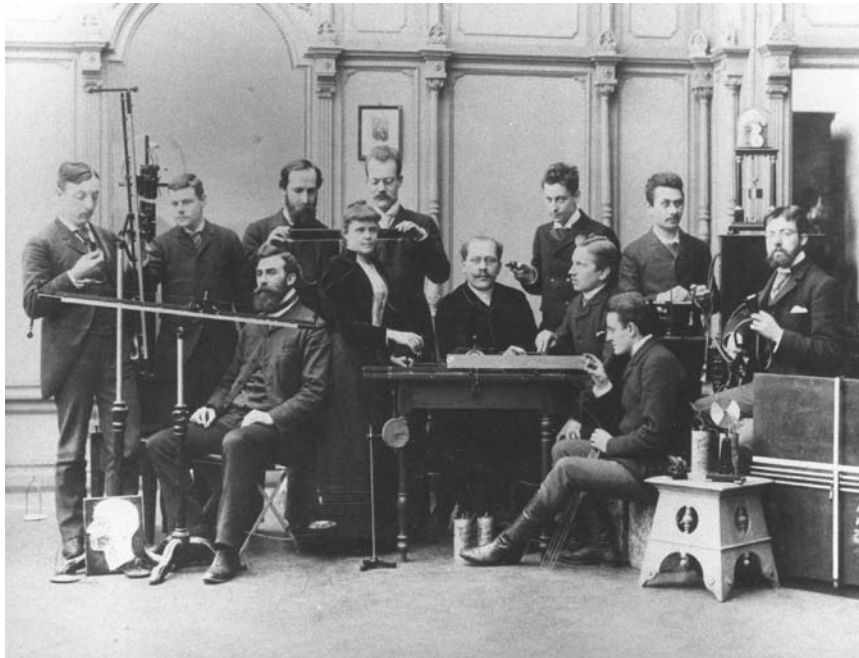
James also had a number of students who went on to play a significant role in the early development of American psychology, even if they did not develop it in a particularly Jamesian direction, such as G. Stanley Hall, who founded the first PhD program in psychology in America at Johns Hopkins; James Rowland Angell (1869–1949), who played a major role in the development of functional psychology at the University of Chicago; Edward L. Thorndike (1874–1949), who established the “law of effect” at Columbia University; and Robert Sessions Woodworth (1869–1962), who promoted a form of functional psychology at Columbia.

Mary Calkins (1863–1930), the first female president of the APA, was also a student of James’s. A Wellesley graduate, she took courses with James, although Harvard refused to formally admit her, despite the support of James and Josiah Royce (1855–1916). She later worked with Hugo Münsterberg on associative memory and defended her dissertation before Münsterberg, James, and Royce, who claimed it was the best PhD thesis that they had ever examined. All this was to no avail, since Harvard refused to award the degree. Calkins worked for most of her life at Wellesley College (where she founded a laboratory) and received numerous awards and honorary degrees. She was elected president of the APA in 1905 and president of the American Philosophical Association in 1918. She never received a Harvard degree and turned down an honorary degree from Radcliffe College in 1902 because she saw it as acceptance of “second best” for women.

Hugo Münsterberg

Hugo Münsterberg accepted the position of director of the Harvard psychological laboratory in 1892. He was one of Wundt's earliest students and research assistants, attaining his degree at Leipzig in 1885 at the age of 22 (followed by a medical degree at Heidelberg in 1887). He took up a teaching position at the University of Freiburg, where he founded a psychology laboratory (at his own expense) and published a number of papers on perception, learning, memory, and attention. Münsterberg disagreed with Wundt on a number of theoretical and methodological issues. He claimed that our apparent consciousness of will is not consciousness of an autonomous act of will that determines behavior, but of an automatic sensational response to our physical anticipation of behavior, based upon muscular feedback. He consequently denied that we have introspective access to acts of will. Wundt refused to accept his work in this area and made him change his dissertation topic to "The Doctrine of Natural Adaptation."

Münsterberg published his theory of the will as *Voluntary Action* in 1888. Wundt and Titchener condemned it but James praised it, which is perhaps not surprising, since Münsterberg's epiphenomenalist treatment of the will matched James's epiphenomenalist treatment of emotion. James met Münsterberg at the



Hugo Münsterberg (center, seated at desk) and students at the University of Freiburg in 1891.

First International Congress in Psychology in Paris in 1889 and cited his writings in *Principles*. James's encouragement (and a personal visit) induced Münsterberg to take up a three-year appointment in 1892. He did not adapt easily to America, and it took some pressure and persuasion to secure his acceptance of a full-time appointment at Harvard in 1897. Wundt consoled him with the thought that, after all, "America is not the end of the world" (quoted in Hale, 1980, p. 55).

Despite the fact that he had to learn English from scratch, Münsterberg became as popular a lecturer as James. He was elected president of the APA and head of Harvard's philosophy department in 1899 and president of the American Philosophical Association in 1907. To a significant degree, he satisfied James's ambition of making the Harvard laboratory the best in America. The archetypal German professor, Münsterberg developed a quintessentially American orientation to psychology as he pursued his academic career at Harvard. Originally hired to develop German experimental psychology, his own interests quickly turned to the practical applications of psychology, and he left the everyday running of the Harvard laboratory to his students Edwin B. Holt (1873–1946) and Robert M. Yerkes (1876–1956).

Popular and Applied Psychology Münsterberg turned out to be a greater promoter of American psychology than most of Wundt's American students and was probably the greatest popularizer of psychology before Watson. He wrote more than 20 books, popular expositions as well as academic texts, and numerous magazine articles on topics such as mental health, education, advertising, and jury trials. James had championed the practical value of psychology, but remained staunchly theoretical and committed to what Münsterberg called "psychic hocus-pocus" such as spiritualism. It was Münsterberg who developed applied fields of psychology such as psychotherapy, forensic psychology, and industrial psychology.

One of the early pioneers of psychotherapy, Münsterberg offered his services free of charge at the Harvard laboratory. He treated people suffering from a variety of disorders, including phobias, obsessions, alcoholism, and sexual problems, based upon his own largely speculative theories and explicitly suggestive therapies. He rejected Freud's theories, although he did acknowledge the sexual etiology of some psychological disorders. He employed hypnosis for a while, until one of his female clients threatened him with a gun, causing the president of Harvard to ban the use of the practice. Münsterberg's book *Psychotherapy* (1909) defined the new field for some years, until his work was eclipsed by Lightner Witmer (1867–1956), the founder of clinical psychology in America.

Münsterberg was also a pioneer in the development of forensic psychology. He demonstrated the variability and unreliability of eyewitness testimony, including the frequently biased and distorted nature of perception and memory. *On the Witness Stand* (1908) was a popular best seller, reprinted in many editions. He questioned the utility of conventional methods of police interrogation

based upon physical force and intimidation and developed physiological tests of veracity—based upon pulse and respiration rates and electrical skin resistance—that were precursors of modern “lie-detector” machines. Münsterberg employed some of these tests in the sensational public trial of Harry Orchard, who confessed to a number of murders but accused the president of the Western Federation of Miners of having ordered them. Based upon the results of what one newspaper called the “lying machine” of “Professor Monsterwork,” Münsterberg declared that Orchard’s testimony was truthful, although the jury acquitted the union boss. While his work was popular with the public, the legal profession scorned it (Wigmore, 1909), and a number of psychologists joined in the critical chorus (including Titchener, 1914). Thus he pioneered but probably also impeded the development of forensic psychology.

Münsterberg was also one of the founders of industrial psychology. He conducted studies of personnel selection, task-oriented aptitude testing, work efficiency, motivation, marketing, sales, and advertising. He published *Vocation and Learning* in 1912 and *Psychology and Industrial Efficiency* in 1913, and his advice was sought by businessmen and government officials. He promoted his views in popular magazine articles and in a movie that played in commercial theatres.

Münsterberg was a colorful and controversial character. His public stance against prohibition, and the financial support he received for it from Adolphus Busch, the German brewer, raised a few eyebrows. Although he supported the PhD candidacy of Mary Calkins at Harvard, he was against graduate education for women and was generally opposed to women pursuing any professional career, including elementary school teaching. He believed that women lacked the capacity for logical reasoning and maintained that they should be barred from juries (and denied the vote). His views were based upon comparative experimental studies of judgments made by men and women in isolation and after participation in group discussion (Münsterberg, 1913b). These studies were precursors of the **social facilitation** studies later conducted by the social psychologist Floyd Allport (Allport, 1920), which explored the influence of social groups on judgment and behavior. Allport studied with Münsterberg, who introduced him to the earlier German work in this area (Meumann, 1907) and suggested it as a dissertation topic.

Münsterberg’s controversial public pronouncements and vigorous popular presentations of psychology alienated many of his academic colleagues, including the more staid faculty members at Harvard. Lightner Witmer claimed that Münsterberg’s self-promotion of his therapeutic success cheapened the profession, and Jacques Loeb called him a “journalistic money-making hack” (Loeb, 1916, cited in O’Donnell, 1985).

Münsterberg developed some influential connections in industry, government, and the arts (including the movie industry). He was friends with Andrew Carnegie and dined with presidents Roosevelt and Taft at the White House. However, none of these connections shielded him from the public odium that followed his

support of Germany's position before and during the First World War. He was vilified by the press and suspected of being a spy. One Harvard alumnus offered the university \$10 million to fire him (which to its credit it did not). At the height of his unpopularity, he made his way through the winter snow to an early morning class at Harvard in 1916 and collapsed from a fatal stroke the moment he entered the lecture hall.

The painting of the philosophy department at Harvard in the early 20th century that hangs in Emerson Hall includes James, Royce, and George Herbert Palmer, but not Münsterberg—although the photograph on which it was based does. In the painting there is an empty chair in the place where Münsterberg stood. According to one story, Münsterberg was painted out after his death (Roback, 1952, p. 208). According to another, he requested to be removed for “aesthetic reasons,” because he was not placed in the center of the painting (Kuklick, 1977).

LADD AND SCRIPTURE AT YALE

The introduction of the new psychology did not proceed quite as smoothly at Yale, where the former Congregationalist minister George Trumbull Ladd (1842–1921) promoted it. Ladd was professor of mental and moral philosophy at Bowdoin College in Maine from 1879 to 1881, during which time he published a number of popular articles advocating the relevance of empirical research in physiology and evolutionary biology to traditional philosophy and theology. To Noah Porter, Yale's president and professor of moral philosophy and metaphysics, he seemed the ideal person to mediate between the old and new traditions in philosophy and psychology (O'Donnell, 1985). Although Ladd had no training in physiology or laboratory experimentation, he began teaching physiological psychology at Yale in 1884. After years of research on physiology and psychology, he published *Elements of Physiological Psychology* in 1887. Ladd's text, like James's later *Principles* that eclipsed it, provided a philosophical justification for the new psychology, as well as a detailed survey of contemporary research, including experimental studies from Wundt's Leipzig laboratory. Yet although *Elements* acknowledged Wundt's contribution, it remained heavily influenced by Scottish common sense psychology, and the later edition (Ladd & Woodworth, 1911) contained only token references to Wundt.

Trouble began when Ladd arranged for the appointment of Edward Wheeler Scripture (1864–1945) as director of the Yale laboratory in 1892. Scripture studied with Ebbinghaus in Berlin and enrolled in Wundt's program in 1888. He completed his PhD degree in 1891 with a dissertation on “Thinking and Feeling.” Scripture was a committed and prolific experimentalist, who initiated the annual *Studies from the Yale Psychological Laboratory* (which ran from 1892 to 1902), albeit largely based upon his own work on sensory tone. When the laboratory attracted

few students, he turned his Leipzig-inspired commitment to precise measurement to industrial research (mainly time-motion studies) and clinical explorations of speech disorders.

Scripture was unwilling to accept Ladd's subordination of psychology to philosophy and criticized the speculative theories of philosophers such as James and Ladd in his popular text *Thinking, Feeling, Doing* (1895). He claimed that the new experimental psychology owed nothing to traditional philosophy, which represented an impediment to the development of the new academic discipline:

And what about philosophy, the science of sciences? Alas! Philosophy is still in the Middle Ages. One by one the other sciences have freed themselves; the lingering clutch of philosophy on psychology is the last hope of respectability.

—(1895, p. x)

He repeated his strident call for the separation of the new psychology from philosophy in *The New Psychology* in 1897. Ladd was so outraged by Scripture's criticism of philosophy and his own work that he demanded that a university committee investigate the conflict within the department. The committee failed to resolve the problem, Scripture was fired, and Ladd was forced to resign. Charles Hubbard Judd (1873–1946), another student of Wundt's, who had been appointed as an instructor in psychology in 1902, took over as director of the Yale program, which took years to recover (Rieber, 1980). Scripture abandoned American psychology and traveled to Britain, where he founded a laboratory in speech neurology in London's West End Hospital for Nervous Diseases in 1912 (O'Donnell, 1985).

HALL AT JOHNS HOPKINS AND CLARK

Granville Stanley Hall was a farm boy from Ashfield, Massachusetts, who made good on his ambition to make his mark in the world. His career was distinguished by a number of psychological firsts. He was the first person to gain a PhD in philosophy at Harvard with a dissertation on a psychological topic (the muscular perception of space). He was the first to create a laboratory and PhD program in psychology in America at Johns Hopkins in 1883. He was one of Wundt's first American students: Although he did not study for a degree with Wundt, he attended some of his lectures and served as a subject in Max Friedrich's laboratory research on apperception in the fall of 1879. He founded the *American Journal of Psychology* in 1887, the first American psychology journal. He founded the American Psychological Association in 1892 and served as its first president. He became the first president of the newly founded Clark University in 1888.

Hall attended Williams College in Massachusetts, where he studied Scottish common sense psychology. Mark Hopkins, professor of moral philosophy and

rhetoric and president of Williams, introduced him to the challenges posed to traditional philosophy by experimental physiology and evolutionary theory. After graduating in 1867, Hall prepared for a career in the ministry at Union Theological Seminary in New York City. He spent much of his time exploring the multifarious delights the city had to offer, such as its theaters, law courts, and zoos. His theological studies suffered, and he came to recognize that he was perhaps not best-suited to the ministry. After he gave his trial sermon at Union, he was summoned to the president's office for the usual critique: Instead of evaluating his sermon, the president fell down on his knees and prayed that Hall might be saved (Hall, 1923).

The Union College faculty encouraged students to study philosophy in Germany (O'Donnell, 1985), so Hall interrupted his studies to travel there in 1869. He took courses in philosophy at the universities of Bonn and Berlin, but also enjoyed the beer halls and other less academic delights of Europe, such as its theaters and zoos. Lack of funds prevented Hall from pursuing a doctoral degree in Germany, so he returned to America, where he completed his degree at Union Theological Seminary in 1870 (although he was never ordained). He worked as a private tutor for a Jewish family in New York City for two years and tried preaching for about three months before accepting a temporary position as Bellows Professor of mental philosophy and English literature at Antioch College in Ohio in 1872 (O'Donnell, 1985). He taught there until 1876, during which time he read Wundt's *Principles of Physiological Psychology*. His passion turned to psychology, which he saw as a "safe route to a permanent philosophical position." He took a leave of absence from Antioch to become an instructor of English at Harvard, where he enrolled as a graduate student with James. Despite his own heavy teaching load, he managed to find some time for physiological research and submitted his dissertation "On the Muscular Perception of Space" in 1878.

After receiving his degree, Hall returned to Germany, where he studied physiology with du Bois-Reymond at the University of Berlin and attended Helmholtz's lectures on physics. He moved to the University of Leipzig in 1879, where he took classes with Ludwig, Fechner, and Wundt. He only took philosophy classes with Wundt, but worked in the Leipzig laboratory during the fall of 1879. He later claimed that he was "on the whole disappointed with Wundt" (Hall, 1923). He returned to the United States in 1880, in debt and with a new wife, but no job prospects.

Johns Hopkins and the New Psychology

Hall recognized early in his career that the new psychology offered the best route for professional advancement in philosophy. He also recognized that the application of psychology to education was an effective means of promoting the fledgling

discipline and advancing his own career within it. He secured an appointment as university lecturer at Harvard, where he gave a popular series of public talks on education in 1881. Their success led Daniel Gilman, the president of Johns Hopkins, to invite Hall to take up an appointment as a part-time lecturer in philosophy and pedagogy at the new university in 1882. Like James with Harvard's Eliot, Hall managed to persuade Gilman that the new psychology could serve as a bulwark against materialism and mechanism and as a buttress for traditional religion. According to Gilman, Hall assured him that the "new psychology, which brings simply a new method and a new standpoint to philosophy, is . . . Christian to its root and center" (Gilman, 1885, p. 48, cited in O'Donnell, 1985, p. 119). Given his religious background and training in physiological psychology (albeit much exaggerated by Hall), Gilman considered him an ideal person to smooth relations between the science-oriented university and the religious establishment and to develop the practical applications of psychology.

Hall was a great success at Hopkins and was elevated to a full professorship in psychology and pedagogy in 1884. He created the first fully functional psychological laboratory in America, reproducing the system of seminars and doctoral certification characteristic of German programs. Joseph Jastrow, the first person to attain a PhD in psychology in America at Hopkins, conducted the first experiments (on perception) in the new laboratory with Peirce (Peirce & Jastrow, 1884) in 1883–1884 (Kihlstrom, 2004).

Hall did not restrict psychology to Wundt's physiological psychology. In his inaugural lecture at Hopkins, he characterized the new psychology as comprising comparative psychology (based upon evolutionary theory), experimental psychology (based upon physiological psychology), and historical psychology (patterned after Wundt's *Völkerpsychologie*). He also staked out his claim that the new psychology provided the scientific basis of education:

Those who devote themselves to the work of education as a profession are strongly recommended to give their chief time and labor to grounding themselves in Philosophy and Psychology, which constitute the scientific basis of their profession. Pedagogy is a field of applied psychology, and if the latter is known the application is not hard to make.

—(1885, p. 248)

His students originally published their experimental studies in Bain's journal *Mind*, but later published in the *American Journal of Psychology*, which Hall founded in 1887 as the first American journal devoted to theoretical and empirical reports in psychology. Jastrow wrote the set of conventions for reporting psychology experiments that later evolved into the "APA style" (Jastrow, 1890). Like many editors of early American journals in psychology, Hall owned the *American Journal of Psychology* and financed it out of his own private income (he sold it to Titchener in 1920). Hall's other doctoral students at Hopkins included John Dewey, Edmund Clark

Sanford (1859–1924), William H. Burnham (1855–1941), and the neurophysiologist Henry Herbert Donaldson (1857–1938).

Clark and Genetic Psychology

In 1888 Hall was invited to join the newly founded Clark University in Worcester, Massachusetts, as its first president, with an academic position as professor of psychology. After a glorious year spent studying the operation of European universities and military academies, Greek archeological sites, brothels, beer halls, circuses, and the inevitable zoos, Hall returned to model Clark as a smaller version of Hopkins (according to the wishes of Jonas Clark, the mining equipment magnate whose endowment established the university). He created five departments—of physics, chemistry, biology, mathematics, and psychology—devoted to graduate studies (although Clark, like Hopkins, was eventually forced to introduce undergraduate programs to maintain the financial viability of the university). Hall left the day-to-day running of the psychology programs to two former students who transferred with him to Clark. E. C. Sanford headed the laboratory and psychology program, and W. H. Burnham headed the program in educational psychology.

While he continued to employ the rhetoric of experimental science in his promotion of psychology as a scientific discipline and profession, by the mid-1890s Hall shifted the direction of research at Clark to **genetic psychology**. According to Hall, whereas experimental psychology was based upon the method of physics and physiology and focused upon measured responses to manipulated sensory stimulation, genetic psychology was based upon the method of evolutionary biology and focused on the development of individual organisms, especially the (extended) development of human children. Hall made genetic psychology the foundation of his developmental and educational psychology.

Hoping to develop a broader institutional base for his work in education (and to secure a share in increased government funding for secondary education, teacher training colleges, and schools of education), Hall formed an alliance with the National Education Association (NEA). In 1891 he founded *Pedagogical Seminary* (later the *Journal of Genetic Psychology*), devoted to the study of child development, and that same year announced the need for scientific pedagogy to an enthusiastic audience at the Toronto meeting of the NEA. Hall became a leader in the emerging child-study movement and instituted a very successful summer school for professional educators in 1892 (which was repeated every year until 1903). He persuaded the NEA that the new psychology constituted the scientific basis of child study; in 1904 it organized a Committee on Psychological Enquiry and its own department of child study. Child study congresses sprang up all over the country, and Hall created the Child Study Institute at Clark in 1910.

Studies of child development at Clark were based upon standard anthropometric measures of sensory acuity and reaction time (originally developed by Galton)

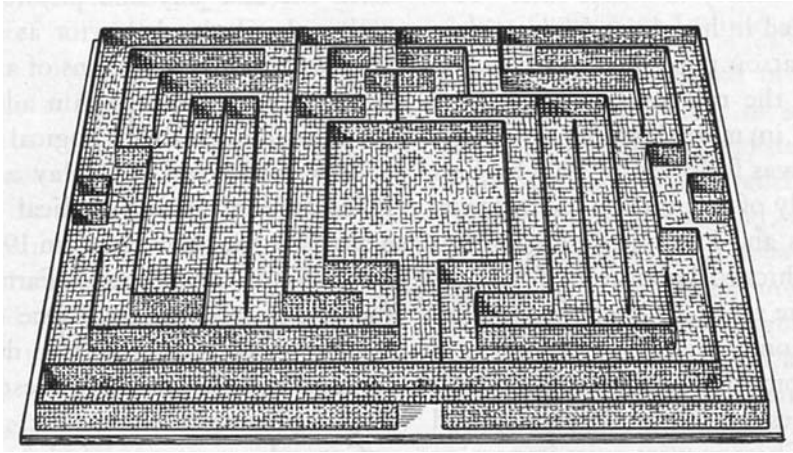
and questionnaires constructed by Hall and his colleagues, which anticipated the later work of Jean Piaget and Lawrence Kohlberg (1927–1987). The projected goal of this genetic psychological research was to establish the course of normal child development so that teaching practices could be suitably adapted to it and to explore the degree to which learning could be facilitated by educational intervention. Hall believed that many differences in children's abilities were innately determined: He was a committed eugenicist who bequeathed a quarter of a million dollars (equivalent to about \$3 million today) in his will for a chair in eugenics at Clark (Rosenzweig, 1984). However, he also believed that child and adult behavior is often the product of adaptive adjustment to the environment.

Most psychologists trained in laboratory techniques were skeptical of the use of questionnaires, and experimental child research at Clark came to grief when Hall's colleague Franz Boas created a public outcry by loosening children's clothes in 1891 (as part of a study of classroom conditions for learning). Animal psychology, which Hall had introduced as an integral component of genetic psychology, provided a convenient solution to both problems. Animal psychology maintained the commitment to objective experimental research while avoiding any suggestion of moral impropriety (although Hall had to defend his animal psychologists against anti-vivisectionist protestors).

Linus Kline (b. 1866), a psychology instructor at Clark, who published *Suggestions Towards a Laboratory Course in Comparative Psychology* in 1899, introduced rats into his laboratory courses in comparative psychology. Adolf Meyer (1866–1950), a physician from Switzerland who took up an appointment at Worcester Hospital for the Insane, and whom Hall invited to teach courses in abnormal psychology at Clark, introduced the albino rat to America. Laboratory rats had many virtues for researchers: They were easy to house and cheap to maintain (unlike cats, dogs, and monkeys) and were much closer to humans than the amoebae and insects that were the focus of early biological studies of animal behavior (Logan, 1999). Kline and his graduate student Willard S. Small (1870–1943) conducted a series of experimental studies of associative learning, in which rats had to burrow through sawdust or gnaw through a door-latch to attain a food reward (Small, 1900). Small also studied maze learning in rats, by constructing a miniature replica of the maze at Hampton Court Palace in London (Small, 1901). These studies, and the work of other Clark students such as Andrew J. Kinnaman (who studied discrimination and intelligence in monkeys), set the stage for the later distinctive focus of animal and human psychology on the problem of learning, although animal psychology at Clark was discontinued at the turn of the century (Boakes, 1984).

The American Psychological Association

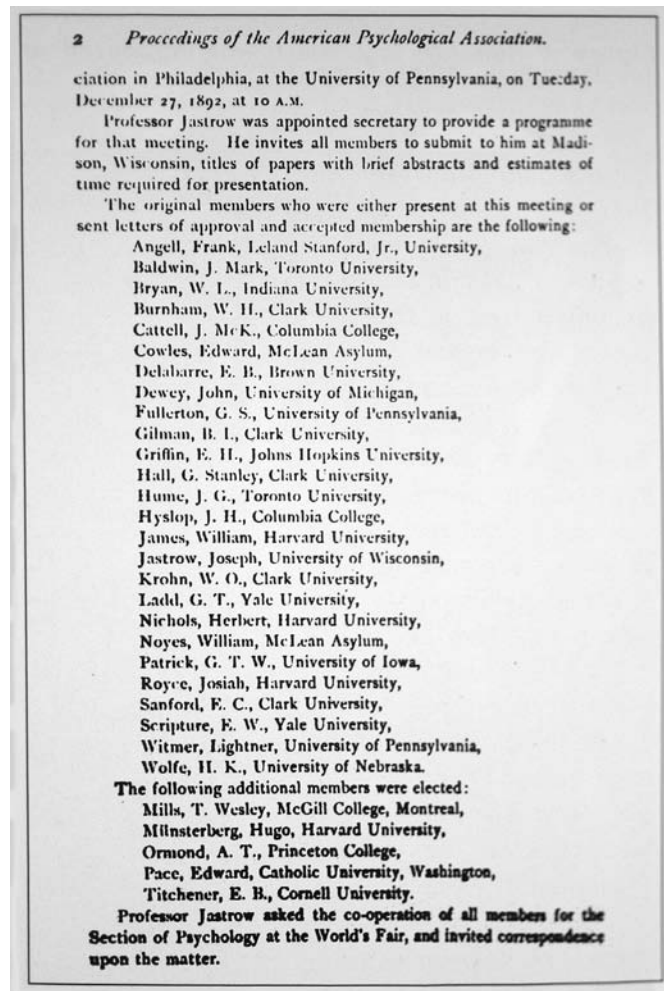
Hall recognized the need for a professional organization for psychologists, both to promote relations with other professional organizations (such as the National



Willard S. Small's replica of Hampton Court maze for the study of learning in rats.

Education Association) and the general public and to fulfill his own personal goal of leadership of the psychological community. In the summer of 1892 Hall invited 26 American psychologists to Clark to form a psychological association. These included prominent philosopher-psychologists such as James and Ladd, but also heads of laboratories and asylums (and 14 of his former students and colleagues at Clark). Only a dozen of the invitees were able to attend the first organizational meeting held in Hall's office on July 8, 1892, where the American Psychological Association was founded, with Hall elected as its first president. Those present heard that many absent invitees had agreed to join (including Dewey, James, Scripture, Witmer, and Harry K. Wolfe), and they elected additional members such as Münsterberg and Titchener. They also agreed to hold their first meeting at the University of Pennsylvania in December 1892, where Hall delivered the first presidential address on the "History and Prospects of Experimental Psychology in America."

From the beginning the APA was inclusive in its membership policy, at least in terms of religion and gender. The charter members included two Jews, Jastrow and Münsterberg, and one Catholic, Edward A. Pace. Two women, Mary Calkins and Christine Ladd-Franklin, were elected members in 1893 (Sokal, 1992). However, the APA was less inclusive in terms of professional qualifications. Despite the common rejection of experimentalism and general advocacy of the practical application of psychology, for many decades membership was restricted to academic psychologists with established publication records. The stated professional aim of the APA was to promote "the advancement of psychology as a science," although exactly what that amounted to remained the focus of intensive debate in the following decades.



List of original members of the American Psychological Association.

Adolescence and Sex

Hall's own developmental research was focused on adolescence, which he defined as ages 15–25. He maintained that adolescence is a critical period of developmental transition, during which humans rely on instincts as they abandon the habits of childhood and begin to embrace those of adulthood. In 1904 he published the two-volume *Adolescence: Its Psychology and Its Relations to Physiology, Anthropology, Sociology, Sex, Crime, Religion, and Education*. Although the work was a best seller (reprinted in many editions), it was not well received by the academic psychological

community, who complained of its emphasis on sex. They were equally shocked by Hall's weekly lectures on sex at Clark (they were officially restricted to a male audience, but he was forced to abandon them because of female gatecrashers and eavesdroppers) and by his revolutionary proposal that sex education be provided in schools. Hall thought that sex played an important role in child development and adolescence, although he did not embrace Freud's psychosexual theories.

Like Münsterberg, Hall was a great popularizer, writing articles on psychology for periodicals such as *Ladies' Home Journal* and *Appleton's Magazine*, usually offering practical advice on dealing with children and adolescents. Although he abandoned his career in the ministry, he maintained an interest in religion, albeit now oriented to the psychology of religion. He created the (short-lived) *Journal of Religious Psychology* in 1904 and in 1917 published a book with the doubtful title *Jesus, the Christ, in the Light of Psychology*, which was not well received.

Despite his free discussion of sexual matters and promotion of sex education, Hall was opposed to coeducation. Since he believed that adolescence is a time of sexual catharsis for men and of preparation for motherhood for women, he thought their institutional physical proximity was a recipe for disaster. Nevertheless, Hall's program at Clark was more hospitable to women and minorities than most other psychology programs of the day. Mary Calkins worked with Edward C. Sanford at Clark in 1890 (although the university did not officially admit women until 1900). Although he endorsed Haeckel's recapitulation theory, which placed blacks at a less developed evolutionary stage, Hall was personally committed to their education. Francis Cecil Sumner (1895–1954), later chair of the psychology department at Howard University, became the first black to be awarded a PhD in psychology in 1920, and Clark graduated more black PhDs than any other institution in the early 20th century.

However, progress in psychology remained slow for blacks. The number of PhDs in psychology increased by only about one per year, reaching a total of 32 by 1950 (Guthrie, 1988). Things were not much better for the first generation of women psychologists, who were forced to choose between family or career (Bolles, 1993; Scarborough & Furumoto, 1987). Prior to the dramatic expansion of professional psychology after the Second World War, those blacks who managed to make a career in psychology found their career trajectories generally restricted to colleges of education (Guthrie, 1988), and women found themselves professionally directed to practical studies of child development (Cahan, 2005).

Hall was influential in shaping the careers of the first generation of American PhDs in psychology: by 1898, he had supervised 30 of the 54 psychology PhDs thus far awarded (R. I. Watson, 1978). Clark produced many graduates committed to Hall's scientifically grounded but practically oriented approach to psychology, and he succeeded in his ambition in establishing the Clark program as one of the top psychology programs, rivaling Harvard in terms of the quality of its faculty and students.



Group photo of psychologists attending Freud's lectures at Clark University.

Front row, left to right: Franz Boas (first), Edward B. Titchener (second), William James (third), William Stern (fourth), G. Stanley Hall (sixth), Sigmund Freud (seventh), Carl G. Jung (eighth), Adolf Meyer (ninth), H. S. Jennings (tenth).

Second row, left to right: C. F. Seashore (first), Joseph Jastrow (second), James M. Cattell (third), Ernst Jones (sixth), A. A. Brill (seventh). Third row, left to right: Sandor Ferenczi (seventh). Fourth row, left to right: E. B. Holt (third), H. H. Goddard (tenth).

Hall invited both Freud and Wundt to be guest speakers at the 20th anniversary of the founding of Clark in 1909. Both declined, citing prior lecturing commitments (and Freud citing the insufficiency of the stipend). However, by changing the date of the anniversary celebrations, and by increasing the honorarium, Hall managed to attract Freud and Jung to give a series of lectures. This was an intellectual coup, for there was widespread interest as well as widespread suspicion of psychoanalysis at the time; and the conference attracted many of the major American psychologists of the day (including James, dying of angina pectoris, who made a great impression on Freud).

Old Age

A powerfully built man (he was over 6 feet tall), full of energy and enthusiasm for all matters psychological, Hall was a personally and intellectually dominating force in the formative years of American psychology. While many of his colleagues

and students found him difficult to get along with, and as enthusiastic in promoting himself as in promoting psychology, most appreciated and respected his multivariied contributions to the discipline.

In his academic career and professional service, Hall achieved his own adolescent ambition to make something of his life. However, his private life and presidency at Clark were not without their heartbreaks and setbacks. Personal tragedy struck in 1890 when his wife and child were killed in an accident, and Hall was stricken with diphtheria. In the early 1890s Clark was attacked in the press over (unfounded) reports of cruel animal experiments. Financial problems caused by Jonas Clark's limited endowment of the university led to (unsuccessful) calls for Hall's resignation by the faculty. He soldiered on, despite his bitterness over president William Rainey Harper's raid of Clark faculty and students for the newly instituted University of Chicago in 1892 (the majority of faculty and about half the student body accepted the generous financial inducements to transfer). Hall resigned as president of Clark in 1919, at the age of 75.

In 1922, at the apt age of 78, Hall published *Senescence: The Last Half of Life*, anticipating the later development of life-span psychology. He published two autobiographies, *Recreations of a Psychologist* in 1920 and *Life and Confessions of a Psychologist* in 1923. In his old age he was critical of the manner in which he saw psychology developing, particularly its fragmentation into schools such as structuralism, functionalism, and behaviorism. In 1924 he was elected to the presidency of the APA for a second time (the second person to receive such an honor after James), but he died later that same year before he could complete his term.

APPLYING THE WUNDTIAN SKELETON: CATTELL, WITMER, SCOTT, AND WOLFE

James, Münsterberg, Ladd, and Hall served as sorts of father (or grandfather) figures of American psychology and occupied the promotional foreground in their lifetimes. Other early American pioneers, many of them students of Wundt's, worked on the establishment of the professional and institutional structure of American psychology and the specialist subdisciplines such as clinical and industrial psychology. While they did not embrace the theoretical details of Wundt's experimental psychology or his *Völkerpsychologie*, they adapted his commitment to scientific objectivity and experiment to their range of distinctively applied interests.

James McKeen Cattell: Mental Testing

James McKeen Cattell (1860–1944) was one of Wundt's first Leipzig students and his first full-time American student. He graduated from Lafayette College (which he entered at the age of 15) in Easton, Pennsylvania, in 1880. His

father, a Presbyterian clergyman, was professor of Greek and Latin and later president of Lafayette College. His mother came from a wealthy family, and Cattell used an inheritance from his grandfather to fund a trip to Europe in 1880. Originally interested in literature, he studied German at the University of Göttingen and studied briefly with Wundt at Leipzig. He did not take part in laboratory work and began preparation of a thesis on the philosophy of Rudolf Hermann Lotze.

An essay on this topic gained Cattell a fellowship at Johns Hopkins in 1882. Under Hall's supervision, he worked on the measurement of the time taken to name colors and objects. Despite his academic promise, Cattell lost the fellowship. This may have been a result of his constant bickering with the president of the university or his youthful experimentation with mental stimulants such as alcohol, nicotine, hashish, and morphine (Sokal, 1981). Hall may also have blocked the renewal of the fellowship because he perceived Cattell as a potential rival (O'Donnell, 1985).

Cattell returned to Leipzig in 1883, with accumulated data, a research design, and plans for the construction of apparatus to complete the experiments he had conducted at Johns Hopkins. He became Wundt's first research assistant, based upon his own recommendation (Boring, 1957). Cattell continued his research on the time taken to perform "cerebral operations," a study well suited to Wundt's own program of mental chronometry. His research on reaction time for the naming of letters was published in Wundt's journal *Philosophical Studies* in 1885; a shorter version titled "On the Time It Takes to See and Name Objects" appeared in the British journal *Mind* in 1886.

Cattell recognized the instrumental value of a Leipzig degree (and of laboratory experience and publications) as a professional credential. However, he did not think much of the quality of the laboratory work at Leipzig, "the work done in it is decidedly amateurish" (Jan. 1885, p. 156, in Sokal, 1981) or of Wundt himself: "Wundt himself is scarcely a great man" (Nov. 1885, p. 193, in Sokal, 1981). Cattell described himself as an "apparatus man," with no interest in theory or philosophy. However, the brash young American and distinguished German professor appear to have got on surprisingly well (and much better than Wundt and Titchener, his professed American disciple). As Cattell wrote to his parents, "Wundt seems to like me and appreciate my phenomenal genius."

Wundt characterized Cattell as representing the "typically American" (*ganz Amerikanisch*) attitude of independence and self-confidence. Wundt allowed him to study individual differences in reaction time, a decidedly Galtonian rather than Wundtian research topic. Cattell studied the psychometrics of attention and individual differences in the effects of fatigue and practice. He created a "gravity chronometer," which enabled stimulus words to be presented at controlled intervals and even managed to convince Wundt to consider the idea of different "styles" of responding.

This was quite remarkable, given the later controversy between James Mark Baldwin and Edward B. Titchener over “type” versus “practice” theories of reaction time (Baldwin, 1895, 1896; Titchener, 1895a, 1895b, 1896a). Wundt, like Titchener, maintained that differences in fatigue and practice are subjective biases to be eliminated or attenuated through experimental control, whereas Cattell, like Baldwin, maintained that such individual differences are a legitimate object of experimental investigation. Cattell’s “Psychometric Investigations” was the first American dissertation completed in the Leipzig program (Benjamin et al., 1992), and he gained his PhD in 1886. He may also be said to have contributed to Wundt’s own published output by introducing him to the typewriter.

After Leipzig, Cattell traveled to England, where he worked with Francis Galton (with whom he had corresponded for some years) and completed the first year of a two-year fellowship at the University of Cambridge. He was hugely impressed with Galton’s work and began to apply the techniques he had developed at Leipzig to anthropometric measurement, which reinforced and refined his interest in individual differences (O’Donnell, 1985). He also supported Galton’s program of positive eugenics and offered each of his seven children \$1,000 as a positive incentive to marry college professors.

When the Seybery chair in philosophy was created by bequest at the University of Pennsylvania, Cattell’s father persuaded president William Pepper that his son would be the ideal person for the position. Pepper may have been swayed by the father’s insistence that salary was not an issue, since Cattell was appointed to the chair in 1887 at an annual salary of \$300 (a paltry sum even in those days, equivalent to about \$6,500 in today’s dollars). When his title was changed to professor of psychology the following year, Cattell boasted that he was the first person promoted to a full professorship in psychology in America.

At the University of Pennsylvania, Cattell initiated a program of mental testing based upon a variety of psychophysical measures, such as grip strength, speed of movement, skin sensitivity, and sensory and motor reaction time. He described these measures in a paper published in *Mind* in 1890, in which he coined the term **mental test**:

Psychology cannot attain the certainty and exactness of the physical sciences, unless it rests on a foundation of experiment and measurement. A step in this direction could be made by applying a series of mental tests and measurements to a large number of individuals. The results would be of considerable scientific value in discovering the constancy of mental processes, their interdependence, and their variation under different circumstances.

—(1890, p. 373)

Cattell continued his program of mental testing when he moved to a more lucrative post at Columbia University in 1891, subjecting hundred of students to

batteries of tests in the hope that psychology might “be applied in useful ways” (1904, p. 185). He assumed that his various tests were measures of intelligence and thus useful indicators of academic performance. However, this assumption was seriously undermined when one of his own students, Clark Wissler (1870–1947), attempted to validate it by exploring the correlation between test scores and course grades (employing Galtonian measures of correlation, including Pearson’s newly developed correlation coefficient). To Cattell’s consternation, Wissler found almost zero correlation (Wissler, 1901). This marked the effective end of the Galtonian program of anthropometric testing in America (although Joseph Jastrow continued it at Michigan for a few years after).

Cattell developed the psychology program at Columbia with great success in the early 20th century. He hired a distinguished faculty, which included former students such as Edward L. Thorndike and Robert S. Woodworth. Columbia became one of the leading programs in psychology, and produced the largest number of PhDs in the early decades of the 20th century (surpassing Harvard, Hopkins, and Clark). Cattell was elected fourth president of the APA in 1895.

His other main contribution to the fledging science of psychology was in his role as editor and publisher. In 1894 Cattell and Baldwin launched *Psychological Review*, which quickly became the premier theoretical journal in psychology and the organ of the APA, publishing its proceedings, news, and presidential addresses. *Psychological Review* was followed by *Psychological Index* (which listed publications in psychology) and *Psychological Monographs* (which published lengthy and specially commissioned experimental and technical reports) in 1895.

Cattell sold his interest in *Psychological Review* to Baldwin in 1904 (after a bitter quarrel) and began to devote more attention to general scientific publications. He owned and edited *Popular Science Monthly*, *American Men of Science*, *School and Society*, *American Naturalist*, and *Science* (which he edited for 50 years). He was the first psychologist admitted to the National Academy of Sciences (in 1901) and served as president of the American Association for the Advancement of Science (which he helped to form) in 1924.

Although his students remembered him with warmth and affection, he gained a reputation as a brash and prickly personality (Sokal, 1971), especially in his relations with college administrators, for whom he reserved a special contempt. A staunch defender of academic freedom and tenure against “academic servitude,” he (along with John Dewey) played a major role in the founding of the American Association of University Professors. Eventually the administrators prevailed, and in 1917 the president of Columbia dismissed Cattell from the university for his pacifism (he had defended his son’s distribution of anticonscription literature in an open letter to the United States Congress written on Columbia stationary). Cattell sued Columbia and was awarded substantial damages, but he never again held an academic position. He continued to edit and publish, and he founded

the highly successful Psychological Corporation in 1921, which began to market psychological expertise to the business community (and continues to do so to this day). Cattell died in 1944.

Lightner Witmer: Clinical Psychology

Lightner Witmer graduated from the Wharton School of Business at the University of Pennsylvania in 1888. After he entered law school, he took a psychology class with Cattell, who persuaded him to take up psychology instead and arranged for Witmer to be his assistant. Cattell urged Witmer to complete his studies of individual differences in reaction time with Wundt in Leipzig. The university promised him a salary hike if he obtained a doctoral degree from Wundt's laboratory, which he did in 1892. Although it served his professional interest, Witmer's Leipzig experience was not happy. Wundt refused to allow him to work with untrained subjects and even refused to allow him to serve as an experimental subject himself, because he did not consider him properly "calibrated."

On his return to the United States, Witmer took over Cattell's position at Pennsylvania (on his recommendation) after Cattell moved to Columbia. Although Witmer later claimed that he got nothing from Germany but a doctoral degree and did not consider that he owed much to Wundt (Witmer, letter to Boring, 1948, cited in O'Donnell, 1985, p. 35), he built up the psychology department and laboratory at Pennsylvania along conventional lines. He began a program of laboratory research based upon his reaction-time studies and published an experimental text titled *Analytic Psychology* in 1902. However, his interests turned increasingly to applied psychology as the years progressed. In 1894 he taught an evening course on child psychology for teachers, which developed into a regular Saturday morning program, and he became interested in learning disabilities and the behavioral problems of schoolchildren. One of his students, Margaret McGuire, an English teacher in the Philadelphia public school system, introduced him to an apparently intelligent 14-year-old pupil with spelling difficulties (Routh & Reisman, 2003). He identified a vision problem that was partially resolved by eyeglasses, and the pupil made some improvement with tutoring (although the case was probably more complex, since the pupil may also have been dyslexic [McReynolds, 1997]). This stimulated Witmer to initiate an informal program of educational intervention, working in conjunction with the teachers and families of the children he treated.

His modest success with largely improvised methods motivated him to institute the Psychological Clinic at the University of Pennsylvania in 1896 and to urge his colleagues at the APA to adopt the "clinical" method (Witmer, 1897). The Philadelphia philanthropist Mary L. Crozer provided funding for *Psychological Clinic: A Journal for the Study and Treatment of Mental Retardation*



University of Pennsylvania Psychological Clinic (staff members Helen Backus, Karl G. Miller, and Alice Jones).

and Deviation, which Witmer founded in 1907 and edited for the next 30 years. In the lead article, he coined the term **clinical psychology** to describe a new diagnostic branch of applied psychology: The term “clinical” referenced the clinical method of medicine from which the diagnostic method was appropriated. In describing the development of clinical psychology from his work with pupils with learning difficulties, Witmer claimed that the ultimate value of psychological science lay in its practical utility:

There is no valid distinction between a pure science and an applied science. . . . The pure and the applied sciences advance in a single front. What retards the progress of one, retards the progress of the other; what fosters one, fosters the other. But in the final analysis the progress of psychology, as of every other science, will be determined by the value and amount of its contributions to the advancement of the human race.

—(1907, p. 4)

Consequently, he maintained that the success or failure of remedial treatment was the best test of the adequacy of a clinical diagnosis.

Witmer’s own clinical program was largely based upon the remedial training of children, or what would nowadays be described as “school psychology,” but he stressed that the clinical method extended beyond the treatment of “mentally and morally retarded children” to the treatment of adults:

The methods of clinical psychology are necessarily invoked wherever the status of an individual mind is determined by observation and experiment, and pedagogical treatment applied to effect a change, *i.e.*, the development of such an individual mind. Whether the subject be a child or an adult, the examination and treatment may be conducted and their results expressed in the terms of the clinical method.

—(1907, p. 9)

Witmer relied upon the help and advice of teachers, social workers, and physicians affiliated with the Psychological Clinic. By 1909 it was staffed by Witmer as director, an assistant director, five trained psychology PhD examiners, a social worker, and three assistant social workers (Routh & Reisman, 2003). The success of his program of clinical intervention and training led to the creation of psychological clinics and training programs at other universities, with new courses introduced in clinical psychology. The Iowa Psychological Clinic and the Clark University Psychological Clinic were founded in 1913, and by 1920 there were psychology clinics at 19 universities (Routh & Reisman, 2003). The expanded Psychological Clinic at Pennsylvania became a major center for the training of clinical psychologists, who went on to staff institutions across the country. Many of these institutions also followed Witmer's initiative in creating residential homes for the treatment of educational and other psychological disorders.

One of Witmer's students was Edwin Twitmyer (1873–1943), a pioneer in speech pathology. Twitmyer worked at the Pennsylvania Psychological Clinic and later became director of the speech clinic. In his doctoral dissertation, he described the classical conditioning of the patellar (knee-jerk) reflex, which he identified independently of Pavlov's work (Twitmyer, 1902/1974).

Witmer was generally skeptical of mental tests and condemned much of the mental testing movement as bad science (although he later became interested in gifted children and defined intelligence as the ability to solve new problems). He devised two psychological tests that came to be widely used as diagnostic tools, the Witmer Formboard and the Witmer Cylinders. While committed to applied psychology, he continued to respect the rigor of the laboratory methods that he had been taught at Leipzig and insisted that applied research should be held to the same critical standards as laboratory research. He roundly condemned James's psychic research and dismissed Münsterberg's psychotherapy as quackery. He was an early advocate of the creation of a society of "serious" experimental psychologists as an alternative to the mongrel membership of the APA, a goal later realized through Titchener's creation of the society known as the Experimentalists.

Unlike later behaviorists who claimed that the results of laboratory experiments on animals such as rats and pigeons could be directly applied to human problems in everyday settings, Witmer maintained that clinical psychology must be based upon case studies of individual persons with particular psychological problems. Indeed, this was the main reason for calling his form of applied psychology "clinical" psychology, because it was based upon the method of case histories associated with clinical medicine (although he had little enthusiasm for the work of Freud, which was also based upon case histories):

I have borrowed the word "clinical" from medicine, because it is the best term I can find to indicate the character of the method which I deem necessary for this work. . . . Clinical psychology . . . is a protest against a psychology that derives psychological

and pedagogical principles from philosophical speculations and against a psychology that applies the results of laboratory experimentation directly to children in the school room.

—(Witmer, 1907, p. 8)

Although his own theory and practice was directed to school and vocational psychology, Witmer played a significant role in establishing the legitimacy of psychological (as opposed to psychiatric or medical) treatments of psychological disorders (McReynolds, 1996); nonetheless, most American clinical psychologists did not devote their attention to psychological therapy until after the Second World War. Witmer remained at the University of Pennsylvania until his retirement in 1937. Before his death in 1956, he was the last surviving charter member of the APA.

Walter Dill Scott: Industrial Psychology

Walter Dill Scott (1869–1955) originally planned for a career as a missionary and gained a divinity degree from McCormick Theological Seminary in Chicago. When he later developed an interest in psychology, he traveled to Leipzig in 1898 to study with Wundt. He received his degree in 1900 with a dissertation on the psychology of impulse. On returning to the United States, he was appointed lecturer in psychology and pedagogy at Northwestern University, where he instituted a psychology laboratory. However, he did little research himself and did not publish any experimental studies. His interests quickly turned to the business applications of psychology, especially advertising.

Scott published *The Theory of Advertising* in 1903, based mainly upon articles written for *Mahin's Magazine* (founded by John Mahin, the head of a Chicago advertising agency), and a spate of articles in periodicals such as *The Woman's Herald*, *Atlantic Monthly*, *Business World*, and *Advertising World*. These were speculative discussions about the association of ideas, suggestion, and individual differences in mental imagery, perception, and attention, which reflected the current “state of knowledge” in these areas (Ferguson, 1962, cited in Benjamin, 1997), although some were based upon research done for businesses on particular problems. Scott published *The Psychology of Public Speaking* in 1906, followed by *Increasing Human Efficiency in Business* and *Influencing Men of Business* in 1911.

Scott is generally recognized as the founder of American industrial psychology (Benjamin, 1997), although Münsterberg was a notable precursor. Borrowing from the work of Galton and Cattell, he created a series of mental tests designed to assess business skills, which he later developed as director of the Bureau of Salesmanship Research at the Carnegie Institute of Technology. He also created rating scales for employee selection, which he adapted for officer selection when



Hawthorne experiment.

he became head of the Committee on the Classification of Personnel in the Army (CCPA) during the First World War (von Mayrhauser, 1989). Scott was awarded the Distinguished Service Medal for his army work in 1919 and was elected president of the APA that same year. He started the Scott Company in 1919 (with former members of the CCPA); it offered counseling services to business. Scott became president of Northwestern University in 1920 and served for 19 years as one of its most successful administrators.

Industrial psychology expanded dramatically after the First World War, especially in the realm of personnel selection and evaluation, but also through studies of industrial efficiency. The most famous of those were conducted at the Hawthorne, Illinois, plant of the Western General Electric Company during the 1920s and 1930s. These studies indicated that changes in lighting and temperature improved efficiency, although investigators later noted that almost any environmental change produced the same effect, as a function of the apparent interest researchers (and by inference managers) were taking in workers' progress (Mayo, 1933). What came to be known as the "Hawthorne effect" was later questioned empirically and criticized for its promotion of the interests of managers over workers (Bramel & Friend, 1981).

Harry Kirke Wolfe: Scientific Pedagogy

Harry Kirke Wolfe (1858–1918), whose parents were schoolteachers, developed a program of research on education conceived of as an aid to “scientific pedagogy.” A native of Nebraska, Wolfe studied memory with Ebbinghaus and took his PhD with Wundt in Leipzig in 1886 with a dissertation on tonal memory (Wolfe, 1886). After a few years as a teacher in California, he was appointed chair of the department of philosophy at the University of Nebraska in 1899, where he instituted a psychology laboratory. Nebraska was one of the land grant universities, and in contrast to Hall at Johns Hopkins, Wolfe failed to secure institutional funding for psychology texts and equipment, because he failed to convince the university of the applied value of experimental psychology. Wolfe adapted the scientific rigor of Wundt’s laboratory to the development of tests of memory, imagination, attention, reasoning, and moral development for use in educational assessment (Benjamin, 1991).

Although Wolfe promoted the value of the new psychology to education as vigorously as Hall, his own lack of training in pedagogy inclined the university authorities to doubt his competence to pronounce on matters of education, and he was dismissed from the university in 1897. After serving as superintendent of the South Omaha Public School System and principal of Lincoln High School (both in Nebraska), he returned to the University of Nebraska in 1906 as professor of educational psychology (O’Donnell, 1985).

EDWARD B. TITCHENER AND STRUCTURAL PSYCHOLOGY

Edward Bradford Titchener was born in Chichester, England. His academic excellence secured him scholarships at Malvern College (an English public school) and Brasenose College, Oxford. He studied philosophy and classics at Oxford, but became a research assistant in physiology in his final year. He developed an interest in the work of Darwin and Huxley and did some physiological studies, which he published in 10 papers in *Nature* between 1889 and 1891 (Tweney, 1997). During this period he read Wundt’s *Principles of Physiological Psychology*, which stimulated his interest in psychology.

Titchener translated the third edition of Wundt’s *Principles* into English. Shortly afterward, Wundt produced the fourth edition, which Titchener also duly translated, only to discover that Wundt had produced the fifth edition! Although he later came to represent himself as the champion of Wundtian psychology in America, there is some question about the adequacy of Titchener’s translations (Blumenthal, 1979), which he claimed rendered “literal” rather than “verbal” translations of Wundt (Creighton & Titchener, 1894). Unlike Charles Judd’s later translations (e.g., Wundt, 1897/1902), Titchener’s translations were not approved by Wundt.

Recognizing that he could not satisfy his new interest in psychology at Oxford or any other British university, Titchener traveled to Leipzig to study with Wundt after completing his degree in 1890 (at the age of 23). He worked with Wundt for two years and attained his doctorate in 1892. Wundt had a huge influence upon Titchener, although they were not close, and Titchener actually saw little of him during his years at Leipzig. He made friends with Oswald Külpe and Frank Angell, an American student who established the psychology laboratory at Cornell in 1891. When Angell accepted a position at Stanford University in 1892, he recommended Titchener as his replacement. Although Titchener was also offered a job at Oxford (where he had been working as a lecturer in biology), he recognized that psychology had no immediate future in Britain and accepted Cornell's offer.

On arrival at Cornell, Titchener immediately set about creating what was to become the largest doctoral program in psychology in America during the early decades of the 20th century. He transformed the original 4-room laboratory into a 26-room laboratory. Between 1893 and 1900 he published *An Outline of Psychology* (1896b), 62 papers, and translations of Wundt and Külpe. For these energetic efforts he was promoted to full professor in 1896. He published *A Primer of Psychology* in 1898, and the magisterial four-volume *Experimental Psychology: A Manual of Laboratory Practice* from 1901 to 1905 (which comprised two manuals for students and two for instructors).

Experimental Psychology became the standard manual for experimental psychology courses in America for the next 20 years (Watson used it in his laboratory courses at Chicago), and it was translated into most European languages and Russian. Although these manuals were eventually superseded, they played a major role in promoting the development of experimental psychology in America and instilling a respect for scientific rigor that came to be shared by even the most virulent critics of Titchener's own experimental program. After 1905, Titchener devoted most of his time to running the psychology program and laboratory and published little research under his own name (Tweney, 1987).

Edwin G. Boring, one of Titchener's students and author of *A History of Experimental Psychology*, described him as the "Englishman who represented the German psychological tradition in America" (Boring, 1957, p. 410). However, he might better be described as the Englishman who presented himself as the representative of the Wundtian tradition in America, albeit successfully for most of the 20th century—until the original Wundt was "rediscovered" in the 1970s (Blumenthal, 1975, 1979; Leahey, 1979, 1981). Titchener modeled the psychology program at Cornell upon Wundt's experimental program in Leipzig, and his own lifestyle upon Wundt's. A popular and effective lecturer, Titchener built the reputation of the psychology program upon his elaborate laboratory demonstrations and the rigorous training he required of his students. He presided over the program in an authoritarian fashion that bordered on the dictatorial, prescribing legitimate projects for PhD dissertations and determining who should work on them (in this

latter respect he was probably more rigid than Wundt). He always lectured in his Oxford gown, which he claimed gave him the right to be dogmatic.

While he was harsh and unrelenting with students who crossed him, he was supportive of those who treated him with the deference he considered his due. He inspired affection and loyalty among his students, who volunteered to wash his car and even took to smoking cigars because they believed that Titchener thought this was an essential habit of a good psychologist (Dallenbach, 1967). He also helped to advance the careers of many female psychologists: During his tenure at Cornell about a third of the PhDs who graduated from the program were women. Although this was Cornell's own policy, Titchener seems to have followed it willingly. He was not miserly in crediting the work of his female graduate students or reluctant to recommend them for academic positions.

Margaret Floyd Washburn (1871–1939), a graduate of Vassar College, was Titchener's first graduate student. She was also the first student to receive a PhD in psychology at Cornell and the first woman to receive a PhD in psychology. She had studied with Cattell at Columbia, but transferred to Cornell (on Cattell's advice) because Columbia did not formally admit women. Washburn's dissertation was on visual and tactile sensation and was later published in Wundt's journal *Philosophical Studies*. She was the second female president of the APA and the second American woman (and first female psychologist) to be elected to the American Academy of Sciences. After teaching at a variety of institutions, she returned to Vassar in 1903, where she set up a psychology laboratory. She remained active in teaching and research until she died in 1939. She published *The Animal Mind: A Text-book of Comparative Psychology* in 1908 (based upon a course she taught at Cornell in 1901) and *Movement and Mental Imagery* in 1916.

Throughout his lifetime Titchener was associated with the *American Journal of Psychology*, founded by Hall in 1887. He served as associate editor from 1895 to 1920, and as editor from 1920 to 1925 (after which his students Karl Dallenbach and Margaret Washburn took over the journal). Most of his own work was published in this journal, because he refused to publish in journals such as *Psychological Review* and *Psychological Bulletin* after he fell out with their editors, Cattell and Baldwin.

Structural Psychology

Like Wundt, Titchener conceived of experimental psychology as the study of immediate experience and maintained that its subject matter, unlike that of the natural sciences, is dependent upon consciousness. However, his program of experimental psychology owed more to the British empiricist tradition of associationist psychology than it did to Wundt's voluntaristic psychology (as Titchener acknowledged in the original preface to *An Outline of Psychology*, 1896b). Titchener's aim was to identify the atomistic constituents of conscious experience and determine the

correlational laws of their combination. His positivistic science was restrictively descriptive, and he eschewed Wundt's appeal to explanatory constructs such as apperception. Titchener called his psychology "descriptive psychology" for the same positivist reasons that B. F. Skinner later called his form of behaviorism "descriptive behaviorism."

Titchener claimed that the primary goal of experimental psychology (at least for the foreseeable future) was to describe the basic structure of the mind: the conscious elements of mind and their modes of combination. For this reason, he characterized his form of psychology as **structural psychology** and distinguished it from **functional psychology**, which he claimed was primarily concerned with the functions of consciousness (Titchener, 1898b, 1899).

The experimental focus of Titchener's program was on conscious elements, defined as the simplest components of consciousness available to introspection that are incapable of further discrimination or analysis. According to Titchener, the experimental psychologist dissects mental experience until he "is left with certain mental processes which resist analysis, which are absolutely simple in nature, which cannot be reduced, even in part, to other processes" (1910, pp. 37–38). He identified these elements as sensations such as sights, smells, and tastes (the elements of perception), images (the elements of ideas), and affections (the elements of emotions).

In addition to Wundt's fundamental properties of quality and quantity (intensity), Titchener added clarity and duration, and extension or extensity (in the case of vision and touch). He claimed that affections have quality, quantity, and duration, but not clarity (since they dissipate when we try to focus attention on them). He rejected Wundt's tridimensional theory of emotion and claimed that the dimensions of high/low arousal and concentrated/relaxed attention are nothing more than combinations of sensation and the affective dimension of pleasant/unpleasant.

Most of the experimental studies conducted in the Cornell laboratory focused on the identification of sensational elements. Titchener claimed to have documented more than 44,435 of these in his 1896 *Outline of Psychology* (which makes good nighttime reading for insomniacs), the majority of which were visual (32,820, with 11,600 auditory). Although the focus of these studies was the introspective analysis of ordinary sensations, some were directed toward more exotic targets and required more invasive forms of experimental intervention. Some experimental subjects reported on sensational and affective elements of urination and defecation (and sex, in the case of married students), while others reported on sensational and affective responses to hot and iced water ingested through a rubber tube they were required to swallow.¹

¹According to Cora Friedline, one of Titchener's graduate students, who recalled these experiments at a talk she gave in 1960 at Randolph-Macon College in Virginia (reported in Schultz & Schultz, 1992, p. 122).

Titchener accounted for the organization of sensory elements into complex perceptual wholes (such as the perception of an apple) by appeal to traditional principles of association, the “laws of connection of the elementary sensory processes,” and combinatory laws of color contrast and tonal fusion. Wundt had supposed that these principles account only for passive forms of perception and cognition and appealed to the principles of psychical resultants and relations to account for more complex psychological processes. Titchener maintained that the principles of association account for all psychological processes and analyzed Wundt’s active attentional processes in terms of differential degrees of clarity among mental elements (analogously, he interpreted feelings of concentration as sensations of muscular responses that accompany vivid sensations).

Inspection and Introspection

If Titchener’s theoretical program was more restricted than Wundt’s, his conception of introspection as an experimental method was broader, although no less rigorous in practice. He repeated Wundt’s mantra of repetition, isolation, and variation as the essence of experimentation: “an experiment is an observation that can be repeated, isolated and varied” (1910, p. 20). However, he relied more heavily on introspective reports than Wundt, who favored objective measures such as reaction time. Titchener’s subjects were specially trained to avoid what he called **stimulus error** (1912, p. 488): They were trained to report only the pure contents of experience, not the (theoretically interpreted) meaning of the sensory array or the purported real-world object of the experience. For example, in reporting their immediate sensory experience of an apple, subjects were required to describe colors, tastes, smells, and the like, but not their theoretically mediated experience: the meaning of the complex or its purported real-world object (the apple).

Like Wundt, Titchener claimed that the processes of introspection and **inspection** (the observation of independent physical objects in the external world) are fundamentally identical, differing only with respect to their objects: The objects of introspection are dependent upon consciousness, whereas those of inspection are not. Consequently he claimed that “in general the method of psychology is much the same as the method of physics” (1910, p. 25). Also like Wundt, Titchener maintained that the validity of introspection as a scientific method, like the validity of inspection, is dependent upon intersubjective agreement among observers about observed properties, whether these be the private objects of introspection or the public objects of inspection. It was the perceived failure to attain intersubjective agreement among introspecting subjects that led to the demise of Titchener’s brand of experimental psychology.

Titchener followed Külpe in calling his method of introspection “systematic experimental introspection.” Although he maintained that there is an “essential likeness” between introspection and inspection, Titchener also acknowledged an important difference that had been the focus of earlier critics of introspection such as Kant and Comte. Attempts to introspect mental processes, unlike attempts to inspect physical ones (with the possible exception of quantum mechanical processes), tend to interfere with the mental processes themselves.

Here, however, there seems to be a difference between introspection and inspection. The observer who is watching the course of a chemical reaction, or the movements of some microscopical creature, can jot down from moment to moment the different phases of the observed phenomenon. But if you try to report the changes in consciousness, while these changes are in progress, you interfere with consciousness; your translation of the mental experience into words introduces new factors into that experience itself.

—(1910, pp. 21–22)

However, Titchener did not think that this difference constituted an insurmountable impediment to experimental introspection. He suggested that one way of alleviating the problem was to delay reporting on mental processes until after their completion, so that in many cases introspection became a form of retrospection or “postmortem” examination. He also maintained that experienced experimental subjects could overcome problems of interference:

The practised observer gets into an introspective habit, has the introspective attitude ingrained in his system; so that it is possible for him, not only to take mental notes while the observation is in progress, without interfering with consciousness, but even to jot down written notes, as the histologist does while his eye is still held to the ocular of the microscope.

—(1910, p. 23)

He conceived of the training of experimental subjects as analogous to the calibration of scientific instruments. He claimed that once experimental subjects (“reagents”) were properly trained in “hard introspective labor,” they found that accurate introspection became a largely mechanical process.

Gestalt psychologists criticized Titchener’s structural psychology for its artificiality and complained that conscious experience cannot be decomposed into isolatable elements. Yet Titchener (like Wundt) was well aware that conscious experience is not a mere aggregation of static elements, but an ongoing process. With James, Titchener recognized that consciousness is like a stream: “our subject-matter is a stream, a perpetual flux, and not a collection of unchanging objects”

(1910, p. 16). He also noted the (Heraclitean) methodological problem this generated for experimental psychology: “In strictness, we can never observe the same consciousness over; the stream of mind flows on, never to return” (1910, p. 19). However, he claimed that subjects can observe sufficiently similar experiential sequences through experimental repetition:

Practically, we can observe a particular consciousness as often as we wish, since mental processes group themselves in the same way, show the same pattern of arrangement, whenever the organism is placed under the same circumstances.

—(1910, p. 19)

He also suggested a means of dealing with the problem, by segmenting sequences of a conscious process (a method that harked back to Aristotle’s studies of the embryology of the chick):

For we must remember . . . that the observations in question may be repeated. There is, then, no reason why the observer to whom the word is called out, or in whom the emotion is set up, should not report at once upon the first stage of his experience: upon the immediate effect of the word, upon the beginnings of the emotive process. It is true that this report interrupts the observation. But, after the first stage has been accurately described, further observations may be taken, and second, third and following stages similarly described; so that presently a complete report upon the whole experience is obtained. There is, in theory, some danger that the stages become artificially separated. . . . In practice, however, this danger has proved to be very small; and we may always have recourse to retrospection, and compare our partial results with our memory of the unbroken experience.

—(1910, pp. 22–23)

***Völkerpsychologie* and Applied Psychology**

Because of Titchener’s vigorous promotion of an experimental science of consciousness, he is often represented as being antipathetic to applied psychology, such as educational and clinical psychology, and to developmental and social psychology, including the form of comparative-historical psychology represented by Wundt’s *Völkerpsychologie*. Yet on numerous occasions Titchener acknowledged their legitimacy. With respect to *Völkerpsychologie*, for example, he affirmed the value of a branch of psychology devoted to the study of the cultural products of different social groups, such as language, myth, and custom (1910, p. 28).

Historians have generally portrayed Titchener as a dismissive critic of Wundt’s *Völkerpsychologie*, based upon a number of negative comments he made in his obituary on Wundt (Titchener, 1921). Yet he maintained an active interest in

the project of *Völkerpsychologie* and was an astute commentator on the methodological problems of any form of comparative psychology that dealt with different social and cultural communities. He was critical of the psychological findings of the Torres Strait expedition, which explored cultural differences in sensory acuity, but was supportive of its intellectual goal (Titchener, 1916). If he was not enthusiastic about the development of *Völkerpsychologie*, it was because he thought it premature until the experimental analysis of consciousness was sufficiently advanced:

The functional psychology of the social mind is, as might be expected, in a very rudimentary condition . . . we must have an experimental psychology of the individual mind, before there can be any great progress.

—(1898b, p. 455)

This was also Titchener's attitude to other branches of psychology, including forms of applied psychology, such as educational and clinical psychology. When he distinguished structural from functional psychology, his aim was not to disparage functional psychology, but to establish the conceptual and developmental priority of structural psychology. He maintained that structural psychology had the best chance of instilling the intellectual rigor that was necessary for psychology to develop as a proper science:

The morphological study of mind serves, as no other method of study can, to enforce and sustain the thesis that psychology is a science, and not a province of metaphysics.

—(1898b, p. 454)

Titchener followed Wundt in characterizing Meumann's "experimental pedagogy" as "educational technology," but never impugned its utility. He recognized the value of abnormal psychology and followed Albert Binet in treating pathological cases as forms of "natural experiment" (1898b, p. 465). He was less charitable toward the work of Münsterberg, whom he accused of "trading science for technology," and programs of mental testing, which he dismissed as "second-rate and cheap" (Titchener, 1914). Yet he was not alone in his critical attitude toward them, which was shared by applied psychologists such as Lightner Witmer and John B. Watson.

Titchener maintained that the experimental analysis of consciousness represented the hard core of scientific psychology and thought that the development of other branches was premature until the scientific core had been firmly established. He also feared (like Wundt, and with similar justification and premonition) that this scientific core was in danger of being diluted and displaced by applied psychology or "technology" (1898b, p. 454).



1916 Princeton Meeting of Experimentalists (Edward B. Titchener in foreground, second from left).

The Experimentalists

Titchener was a charter member of the APA, but never attended its meetings, even when they were held at Cornell. He resigned from the organization in 1904, ostensibly because of its failure to act on a charge of plagiarism made against Edward W. Scripture, relating to the unacknowledged use of his translation of Wundt's *Principles* in Scripture's *Thinking, Feeling, and Doing* (1895). That same year, Titchener founded his own group, which came to be known as the **Experimentalists** (reconstituted in 1929 as the "Society of Experimental Psychologists"). Titchener ran this group—from 1904 to 1927—in the same authoritarian fashion as he ran his PhD program and determined the legitimate topics and attendance. This led to the public humiliation of Gordon Allport (1897–1967), the social psychologist and personality theorist, when the Experimentalists met at Clark University in 1922. Allport, who was a graduate student at Harvard at the time, presented a three-minute paper on personality traits. This was met with a blank silence, "punctuated by a glare of disapproval from Titchener" (Allport, 1967, p. 9), after which the group pointedly continued their discussion of sensory perception.

Titchener ran the group like a gentleman's club, inviting only the top people from the top laboratories. Robert Woodworth of Columbia was blacklisted for not

having behaved like a gentleman. Unable to attend, he had improperly posted his personal invitation on the Columbia psychology department notice board with the comment “Who wants to go?” (Boring, 1938). Women were excluded during Titchener’s lifetime, and female faculty and students from Bryn Mawr College were “promptly turned out” when they tried to attend the 1907 meeting at the University of Pennsylvania (Titchener, letter to Münsterberg, 1908, quoted in Furumoto, 1988). In later years they hid under the tables and listened at the doors. Christine Ladd-Franklin (1847–1930), the Vassar graduate who studied with Helmholtz and Müller, waged a long campaign to have women admitted. When Titchener tried to defend his exclusion of women by appeal to the need for critical discussion and cigar smoking, she responded that she was as critical as any man and enjoyed cigars as much as any man.

Titchener dominated the Experimentalists during his lifetime. However, the invited membership was fairly eclectic and included comparative, behaviorist, and clinical psychologists such as Robert M. Yerkes, John B. Watson, and Lightner Witmer. Although few accepted Titchener’s own brand of structural psychology, many early American psychologists felt the need for a more scientifically and experimentally oriented alternative to the APA. In 1898, Witmer proposed the idea of an experimental society that would “exclude half-breeds and extremists” (Hall, quoted in Furumoto, 1988, p. 96). Titchener opposed it because of potential conflict with the APA, but the type of society he described in the form letter advertising the Experimentalists in 1904 was almost identical to that originally suggested by Witmer.

Imageless Thought

Titchener’s program of structural psychology came to methodological grief over the debate about “imageless thought.” As noted earlier, a number of researchers at the University of Würzburg claimed to have detected thoughts that were not accompanied by images (Kulpe, 1912/1964). Binet in Paris and Woodworth in New York reported similar results at around the same time (Binet, 1903; Woodworth, 1906). Wundt had responded critically to the Würzburg studies (Wundt, 1907), but his critique was focused on the method of “systematic experimental introspection” and not on the results obtained (with which he in fact agreed). Wundt was equally critical of Titchener’s own method of “systematic experimental introspection” (Wundt, 1900), even though Titchener insisted upon the rigorous training of experimental subjects (which the Würzburg psychologists did not).

Titchener took issue with the Würzburg results as well as the experimental procedures, which he condemned as sloppy. Between 1907 and 1915 he repeated the Würzburg experiments with his own students at Cornell. He claimed that in all cases subtle sensations and images could be identified by “properly trained observers” in controlled experiments (Clark, 1911). Woodworth, who had spent a

summer working with Külpe at the University of Bonn in 1912, defended imageless thought in his presidential address to the 1914 meeting of the APA at Yale. At that same meeting, John Baird tried to support Titchener by arranging a public demonstration of the effectiveness of systematic experimental introspection. Unfortunately, it was a public relations disaster, which succeeded only in boring the audience to tears (Boring, 1953).

Wundt and Titchener had stressed the fundamental identity of inspection and introspection. They maintained that the scientific objectivity of inspection of the properties of physical objects (their size and acceleration, for example) is grounded in intersubjective agreement among observers, which is a necessary condition for any empirical science. Wundt and Titchener held that the introspective study of immediate experience could be objective because they believed that intersubjective agreement could be attained via laboratory control and repetition. Yet the imageless-thought debate raised serious doubts about this, since the Würzburg and Cornell labs produced consistently different results.

Titchener treated the ability to report images for thoughts as the criterion of a “properly trained observer” and dismissed the recalcitrant results of other laboratories as due to subject naiveté or insufficient training. In this, he may have been influenced by his own experience as a student of Wundt’s. Witmer recounted how Wundt had made Titchener “do over again an investigation . . . because the results obtained by Titchener were not as he, Wundt, had anticipated” (Witmer, letter to Boring, 1948, quoted in O’Donnell, 1985, p. 35). Titchener’s attitude naturally led to the suggestion that the “reagents” in his laboratory were subject to a form of experimenter bias (Müller, 1911–13). Ogden (1911) attributed the different results of the Cornell and Würzburg laboratories to “unconscious bias” due to the different forms of training that subjects received in them. He also suggested that if Wundt’s critique of the form of “systematic experimental introspection” employed by Külpe and the other Würzburg researchers was sound, it could be generalized to all methods of introspection, including those employed by Wundt and Titchener (Humphrey, 1951; Woodworth, 1938). Consequently, many psychologists concluded that the imageless-thought debate was incapable of empirical resolution by appeal to introspective reports.

The Eclipse of Structural Psychology

Yet in a sense the imageless-thought debate was a side show. Cornell remained a bastion of structural psychology during Titchener’s lifetime, and his laboratory texts continued to be employed to instill respect for scientific rigor in psychology students. However, Titchener became increasingly isolated in the first two decades of the 20th century. Critics such as Cattell (1904), Thorndike (1905), Judd (1907), Pillsbury (1911), Dunlap (1912), and Dodge (1912) raised

doubts about the utility of introspective methods, so that by the time Watson launched his famous attack on introspection in "Psychology as the Behaviorist Views It" (1913), he was preaching to a largely converted audience. Moreover, the theoretical and methodological debate about introspection lagged behind experimental practice. By 1898, only about 2 percent of the published experimental studies in psychology in America employed introspection, and only about half employed adult human subjects (Bruner & Allport, 1940, cited in O'Donnell, 1985).

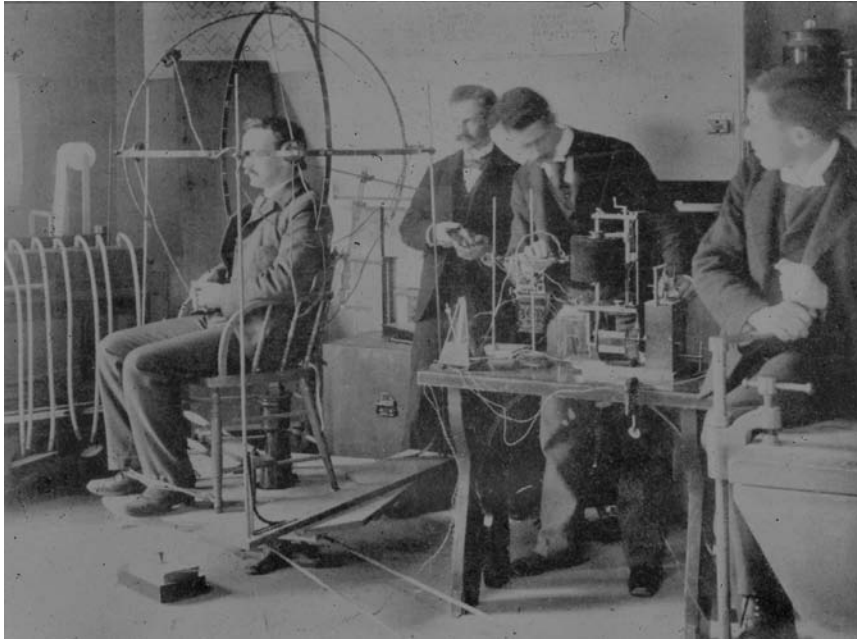
In later years Titchener became somewhat of a recluse. Although a legendary figure at Cornell, some of his colleagues never met him. After 1909, he lectured only on Monday evenings and worked mostly from home, with his wife screening his telephone calls. He conducted a musical group from home on Sunday evenings. He spoke half a dozen languages, including Arabic and Chinese, in order to authenticate his extensive coin collection. He died of a brain tumor in 1927 at the age of 60 (his pickled brain remains on display at Cornell).

When Titchener died, structural psychology died with him. The Experimentalists continued to meet, although they no longer restricted their research topics, or their membership to men. Madison Bentley (1870–1955), a loyal Titchener supporter and founding member of the Experimentalists, took over the chairmanship of the Cornell department. Within a few years, he had transformed it by introducing new specialties such as educational and clinical psychology.

SCIENTIFIC AND APPLIED PSYCHOLOGY

The American students who studied with Wundt returned to America impressed by the scientific rigor of Wundt's experimental laboratory, but developed their own vision of the discipline as they strove to develop more applied and socially relevant forms of psychology. This reorientation was manifest in the development of the APA. The founding meeting held at Clark University on July 8, 1892, had a distinctly Wundtian flavor. Many of the experimental papers that the Clark faculty and visitors presented were on topics in physiological psychology. Yet the first annual meeting of the APA at the University of Pennsylvania in December 1892 reflected the growing eclecticism and applied orientation of American psychology. Cattell and Witmer gave papers critical of psychophysics and mental chronometry. William Brian discussed the use of psychological tests in Indiana schools. Herbert Nichols reported experiments on hospital patients undergoing physiological examination of pain. Joseph Jastrow described the laboratory demonstrations he planned to set up for the psychology exhibit at the Chicago World's Fair in 1893 (Sokal, 1992).

As the century developed, the APA continued to grow, although it went through a lean period when the philosophers broke away to found their own professional



Experiment on influence of dizziness on localization of sounds, presented by Joseph Jastrow at the Chicago World's Fair (1893).

association. It also suffered temporarily as a result of Titchener's formation of the Experimentalists. The 12th annual meeting of the APA in 1902 was held in a school hall in Princeton and was poorly attended. There were only 12 papers delivered, in contrast to 26 the previous year (Furumoto, 1988). By 1910 the membership had expanded from the original 31 charter members to 228, and to 1,113 by 1930 (Fernberger, 1932). Yet many of the 1930 membership were from the new class of nonvoting associate members, founded in 1926 to accommodate psychologists without academic institutional affiliations (Pickren & Fowler, 2003).

DISCUSSION QUESTIONS

1. The academic disciplines of sociology and anthropology were founded in America at around the same time as psychology, yet psychology was much more successful with respect to its institutional development than sociology and anthropology. Why do you think this was the case? Why was the emerging American university system so hospitable to psychology?
2. Why do you think that William James remains such a popular figure in psychology despite his limited contribution to it?

3. When he defined clinical psychology, Witmer claimed that it was a diagnostic branch of applied psychology based upon case histories. Is it still today? Should it be?
4. Harry Kirke Wolfe was fired from the University of Nebraska because educators objected to having someone pronounce on education who had never received training in pedagogy. Is it reasonable to require that educational psychologists receive teacher training? Should industrial psychologists have experience in industry?
5. Is introspection fundamentally the same as inspection, as Titchener claimed? Or are they fundamentally different? If intersubjective agreement cannot be attained over the issue of imageless thought, does that mean that introspection should be completely abandoned in psychology?

GLOSSARY

analytic philosophy Form of 20th-century Anglo-American philosophy based upon the conceptual analysis of abstract systems of thought.

clinical psychology Term coined by Lightner Witmer to describe a “diagnostic” branch of applied psychology based upon the case study method of medicine.

Experimentalists Group of experimental psychologists founded by Titchener in 1904 as a more scientifically rigorous alternative to the APA.

functional psychology According to Titchener, the form of psychology concerned with the functions of consciousness.

genetic psychology Form of developmental psychology based upon the method of evolutionary biology.

inspection The observation of independent physical objects in the external world.

James-Lange theory of emotion Theory that emotion is not the cause of physiological arousal and behavior but the experience of physiological arousal and behavior.

mental test Term coined by Cattell to describe psychophysical measures.

Metaphysical Society Harvard society founded in the 1870s devoted to the discussion of philosophical problems of the day.

new psychology Term employed by early American psychologists to characterize experimental psychology.

phenomenological philosophy Form of 20th-century Continental European philosophy concerned with the abstract essence of thought.

pragmatism View that the adequacy of any theoretical system should be judged by its practical utility.

pragmatist theory of meaning (Peirce) View that the content of a belief or proposition is specified in terms of its empirical consequences.

pragmatist theory of truth (James) View that a belief or proposition is true if it works to our satisfaction.

pragmatist philosophy Form of 20th-century American philosophy that was an offshoot of evolutionary theory and American functionalist psychology.

psychologism Theory that logic and conceptual relations can be treated as naturalistic psychological “laws of thought.”

social facilitation Influence of social groups on judgment and behavior.

stimulus error According to Titchener, the error of confusing the pure contents of experience with the meaning of the sensory array or the real-world object of experience.

structural psychology According to Titchener, the form of psychology that aims to describe the basic structure of the mind: the conscious elements of mind and their modes of combination.

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Functionalism, Behaviorism, and Mental Testing

IN THE EARLY DECADES OF THE 20TH CENTURY, WUNDT'S STUDENTS AND the indigenous pioneers of American psychology continued to promote and develop their varied conceptions of scientific psychology. Cattell claimed in his 1895 presidential address to the APA that the "wide range of individual interests" of American psychologists demonstrated their "adjustment in a complex environment" (1896, p. 134). Yet these different interests also proved to be forces of division in the following decades, sometimes characterized as the period of the competing "schools" of psychology (Murchison, 1930; Woodworth, 1931).

Titchener's form of structural psychology remained an intellectual force in the first two decades of the 20th century, but became increasingly isolated at Cornell and was eventually displaced by functionalist and behaviorist psychology. As the 20th century developed, psychology in America distanced itself from its philosophical roots, including the notion that scientific psychology should be grounded in the introspective analysis of consciousness. Many psychologists had abandoned introspective methods by the time John B. Watson issued his behaviorist "manifesto" in 1913 (Watson, 1913a). This was partly because of the "imageless thought" debate, but also because they had little use for introspection in their applied educational, industrial, and clinical work. Psychology in America also distanced itself from its roots in German physiological psychology. Although courses and demonstration practicals in laboratory methods continued to be employed in the PhD certification of American psychologists as bona fide practitioners of scientific psychology, the "brass instruments" of the new psychology were increasingly appropriated and adapted by educational, industrial, and clinical psychologists as part of their battery of mental and physical aptitude tests.

Many psychologists who came to embrace Watson's (1913a) call for a behavioral science of prediction and control did so because it suited their already well developed applied interests, not because they were convinced by Watson's arguments or rhetoric. These interests were frequently a product of the various institutional and social pressures and opportunities that promoted the development of